Blavatnik Index of Public Administration 2024: Methodology

A guide to the principles, methods and sources used to produce the Blavatnik Index of Public Administration 2024

Blavatnik School of Government

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Preface

This is a replica of the methodology section of the Blavatnik Index of Public Administration website, the documentation on the website remains the definitive source. This replica has been produced in Quarto with the purpose of to integrating the documentation into a single PDF document, an epub version is also available.

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Media contact: comms@bsg.ox.ac.uk

All other enquiries: blavatnikindex@bsg.ox.ac.uk

1 Background

The Blavatnik Index of Public Administration 2024 is a tool designed to provide leaders of national civil services and public administrations with a clear, accessible and robust dataset on how civil services and public administrations around the world compare. Our ambition for the Index is to foster a more data and evidence informed approach to the management and reform of civil services and public administrations.

The Index builds on the Blavatnik School of Government's previous experience of collaborating, with the Institute for Government and the UK Cabinet Office, in the production of the International Civil Service Effectiveness (InCiSE) Index from 2016 to 2020. While inspired by and a conceptual successor to the InCiSE Index, changes to methodology, data availability and country coverage mean that the results of the Blavatnik Index of Public Administration are not directly comparable to the results of the InCiSE Index.

1.1 The need for an index of public administration

Benchmarking is a common practice in performance management, the comparison of a unit (a company, a school, an employee) with its peers can identify that unit's strengths and weaknesses and potential opportunities for improvement. As a result of its natural monopoly, it can be difficult to compare the activities and characteristics of a country's national government with other organisations and institutions within that country. Therefore, international comparisons with how other governments operate are often sought.

There already exist many sources of data for international comparisons, which can be difficult for the leaders of civil services and public administrations to distil and understand how their country is performing. The purpose of the Blavatnik Index of Public Administration is to provide a coherent approach to using these existing data and to focus on the data most relevant to the operation of civil services and public administrations. Table 1 1 provides a comparison of the Blavatnik Index with other types of data available to officials for comparing how their civil service or public administration compares with others.

The principal benefit of the Blavatnik Index of Public Administration is a tool that supports peer learning, by providing comparative analysis of data that senior officials and ministers that measures things they can directly act on.

In the achievement of this goal the Index should be used to complement other sources of data and evidence in any exercise to understand how a country's civil service or public administration is performing. In addition to other types of international comparisons (e.g. measuring economic, social or environmental outcomes), governments will have their own set of performance targets and outcome data that relates to their specific political goals and takes account of their particular context.

Table 1.1: Comparison of other types of data sources for international benchmarking of national governments and public administrations

Types of comparative source	Comment	Examples
Governance indices	In addition to assessments of the executive branch of government these indices tend to include assessments of: the legislative and judicial branches; the nature of democracy and political institutions; media, academic and civil society freedoms; and, possibly, the achievement of policy outcomes. Many of these additional aspects are outside the direct influence of officials and are the result of political choices/cultures.	• World Bank Worldwide Governance Indicators • Bertelsmann Transformation Index • Sustainable Governance Indicators • Chandler Good Government Index
Thematic indices	These indices focus on a single topic of interest. As a result, they either focus on a narrow aspect of the activities of government or only a small subset of its data directly measures activities or characteristics of civil services and public administrations.	• Transparency International Corruption Perceptions Index • World Justice Project Rule of Law Index • OECD iREG Index
Regular data collections	Like thematic indices these data collections often (but not always) focus on a single topic of interest, however they may have no (or multiple) headline measures rather than a single index. As with thematic indices they either focus on a narrow aspect of government or only a subset of its data is directly relevant to the management and leadership of public administrations.	• European Commission eGovernment Benchmark Report • UN’s Sendai Framework Monitoring • Quality of Government Institute Expert Survey • International Survey of Revenue Administration

2 Conceptual framework

In order to create an Index, a framework is needed that we can use to conceptually describe the activities and characteristics of a public administration and to align source data against. Our intention is for the framework to not only provide a theoretical underpinning to the Index but also ensure it is usable by practitioners. The conceptual framework for the Blavatnik Index of Public Administration builds on that developed for the InCiSE Index, but based on desk research and feedback from academic and practitioner stakeholders it adopts a restructured and expanded framework.

Figure 2.1: The Blavatnik Index of Public Administration's conceptual framework

The core focus of the Index are the activities, characteristics and outputs of public administrations. These are organised into four domains that provide a high-level structure for thinking about the activities of public administrations:

- **Strategy and Leadership** – measuring activities that steer and coordinate the functioning of the public administration as well as the characteristics and values that support the stewardship of the institutions of the public administration.
- **Public Policy** – measuring the core executive activities of public administrations in support of the country's national government.
- **National Delivery** – measuring public services delivered directly by national governments and the oversight of other systems of public service delivery.
- **People and Processes** – measuring the activities and characteristics in support of the public administration's workforce.

In addition to the core components that are measured by the Index, our framework also identifies the aspects about the wider context which public administrations are situated within but are not measured by the Index.

Civil services and public administrations exist to execute the practical work of government, they take inputs (political direction, public finances and human resources) and turn them into outcomes and impact (economic, social and environmental changes). Public administrations also operate within their own national context (constitutional and legal frameworks; national political climate and debate; economic, social and environmental conditions; and, international context and actors) which can influence both the inputs they have available to them as well as their capacity to act or scope for action. There are also other actors (sub-national governments and other public service entities; businesses and civil society actors; communities and individual citizens) who influence the outcomes and impact of public policy and service delivery.

2.1 Strategy and Leadership domain

The Strategy and Leadership domain seeks to assess the setting of the strategic direction for the government's programme of work, the stewardship of public institutions, and the overarching values that guide the behaviours of and approach taken by public officials. It is made up of five themes: strategic capacity; cross-government collaboration; openness and communications; integrity; and, innovation.

2.1.1 Strategic capacity

The strategic capacity theme seeks to measure the ability of the centre of government to set strategic direction and to ensure that the institutional structure of government remains fit for purpose. Centres of government support the political leader(s) of a national government and as such need to both set out a vision that other ministries and agencies can follow as well as ensure organisations have the resources and capacities to achieve the objectives they are tasked with (Barber 2016).

2.1.2 Cross-government collaboration

The cross-government collaboration theme seeks to measure the extent to which different parts of the national government work together in the development of policy or the delivery of services. In particular it aims to focus on voluntary collaboration, i.e. where ministries and agencies chose to work together in pursuit of a common goal (e.g. setting up a joint project team) rather than where they are mandated to work together as a result of other processes (e.g. debating policy proposals in a cabinet committee). Modern policy challenges are increasingly cross-sectoral in nature, necessitating multi-agency or whole-of-government responses (Verschuuren, Hilderink, and Vonk 2020). Such multi-agency collaboration must be more than formal committee processes, successful multi-agency collaboration fosters collective focus on the government-wide object rather than the narrower interests of specific ministries or programmes (Brown, Kohli, and Mignotte 2021).

2.1.3 Openness and communications

The openness and communications theme seeks to measure: the extent to which governments consult with citizens and stakeholders in policy development; the extent to which laws, regulations and government information is publicly available. The World Bank Group (2017) notes that "transparency initiatives [are] an important first step toward increasing accountability". Ball (2009) summarises the uses of transparency in three metaphors: (i) 'a public norm to counter corruption'; (ii) open government and decision making; and, (iii) transparency via evaluation of policy actions and programmes.

2.1.4 Integrity

The integrity theme seeks to measure the extent to which public officials make decisions and exercise their duties impartially and do not engage in corruption; integrity is often identified as a core value associated with public service. The International Civil Service Commission (2013) highlights the importance of integrity to the work of the United Nations (UN) common systems staff: "The concept of integrity ... embraces all aspects of behaviour of an international civil servant ... including ... honesty, truthfulness, impartiality and incorruptibility. These qualities are as basic as those of competence and efficiency". Many studies of good governance have used measures of integrity in their analyses, for instance Muriithi et al. (2015).

2.1.5 Innovation

The innovation theme seeks to measure the degree to which new ideas, policies, and ways of operating are able to develop freely. The importance of innovation for public sector organisations has been identified for over decade (OECD 2015b). Mulgan (2014) argues that innovation needs to be treated 'with the same seriousness they deal with handling risk, financial controls or regulatory enforcement'.

2.2 Public Policy domain

The Public Policy domain seeks to assess the core public administration functions, the activities that are fundamental for any national government. It is made up of five themes: policy making; financial management; regulation; crisis and risk management; and, the use of data.

2.2.1 Policymaking

The policymaking theme seeks to measure the extent to which governments can develop effective policy. The formulation and implementation of sound policy is an essential function of government (Kaufmann, Kraay, and Zoido 1999). The OECD (2012) note the importance of high quality policy development and advice as a core requirement for good governance.

2.2.2 Financial management

The financial management theme seeks to measure the extent to which governments manage public money effectively. Holt and Manning (2014) note that fiscal and financial management is one of the central management systems of public administrations. While the mere adoption of a fiscal rule does not guarantee better fiscal performance, the latest cross-country analysis finds that well-designed fiscal rules are associated with lower government deficits, debts, and borrowing costs (Hughes, Leslie, and Pacitti 2019).

2.2.3 Regulation

The regulation theme seeks to measure the use of impact assessment in the development of regulations and that regulations are enforced properly and efficiently. The OECD (2012) "recognis[es] that regulations are one of the key levers by which governments act to promote economic prosperity, enhance welfare and pursue the public interest" and that "well designed regulations can generate significant social and economic benefits which out weigh the costs of regulation, and contribute to social well-being". While Torfing and Triantafillou (2016) argue the importance of the collaborative development of regulation with citizens, proactive regulation can be a lever for social change, Aksoy et al. (2020) note that regulations legalising same-sex partnerships/marriages have led to increased acceptance of homosexuality in Europe.

2.2.4 Crisis and risk management

The crisis and risk management theme seeks to measure how well governments prepare for and manage critical risks to the functioning of their country's society and economy. Baubion (2013) highlights crisis management as central to government's role and a "fundamental element of good governance". Christensen, Fimreite, and Lægreid (2011) discusses how credibility and trust in governments to deal with crises is vital both to reassure and encourage support from the private sector and general public.

2.2.5 Use of data

The use of data theme seeks to measure the extent to which governments have the data, information and skills necessary to develop policy and deliver public services. The importance and role of evidence-informed policy making is long understood (Langer, Tripney, and Gough 2016), the effective use of management information is also critical for improving the performance and operation of front-line services (Brown, Kohli, and Mignotte 2021).

2.3 National Delivery domain

The National Delivery domain seeks to assess the ability of the national government to oversee the delivery of public services, including those services it delivers itself. It should be noted that the responsibility and nature of public sector delivery varies considerably between jurisdictions, the Index does not and cannot seek to be a comprehensive comparative assessment of all types of public service delivery. Deliberately, the Index has sought to avoid services which are highly varied in their constitutional/operational arrangements and/or where their delivery is highly tied policy goals (e.g. health, education, labour market). This domain is made up of five themes: system oversight; digital services; tax administration; border services; and, social security.

2.3.1 System oversight

The system oversight theme seeks to measure the extent to which the government can achieve its policy objectives through its own means and through leadership and stewardship of wider delivery systems. Centres of government have an important role in holding ministries, agencies and others involved in the achievement of policy goals, to account and monitor progress (Barber 2016), effective central units also use their monitoring and accountability functions to collaboratively support and assist line ministries to achieve their objectives (Brown, Kohli, and Mignotte 2021).

2.3.2 Digital services

The digital services theme seeks to measure the government's support for digital public services through the strategies and policies that support their development, the technologies that enable them to work effectively, and the end user experience. As identified by Sullivan et al. (2021) the digitization of services has become mandatory rather than preferable, because digital services can offer sizeable benefits in the speed, reliability, accuracy, and convenience of service provision (Hinkley 2022).

2.3.3 Tax administration

The tax administration theme seeks to measure the operational quality of a country's national level tax administration. Fukuyama (2013) argues that taxation is a "necessary foundation of all states", while Holt and Manning (2014) highlight the importance of tax administration in measuring the effectiveness of public administration.

2.3.4 Border services

The border services theme seeks to measure the operational quality of a country's national borders, the extent to which legitimate goods/services can be transacted across the border and ease with which tourists and business visitors can enter/leave the country. The IMF (2012) notes that customs are often a bell-weather for the degree of corruption and arbitrariness in the public sector, and that effective reform can help improve perceptions of a country's institutions. The adoption of "single-window" systems for customs operations are seen as crucial development in improving the efficiency of customs operations (UNECE 2020). International migration (i.e. the movement of people to move for medium- to long-term periods) is excluded as this is closely associated with policy intentions, with some countries having policy to explicitly encourage in-migration while others try to limit either in-migration or out-migration.

2.3.5 Social security

The social security theme seeks to measure the operational quality of a country's national social security system. Chalam (2014) notes that social security systems have "played an important role to alleviate poverty and provide economic security" in all successful societies and economies, moreover McKinnon (2011) notes that "several legal instruments adopted by the United Nations recognised social security as a basic human right".

2.4 People and Processes domain

While the first three domains focus on the operational functioning of the public administration, in effect "what" public administrations do or "how" they do it, the People and Processes domain seeks to assess what and how it feels to work in or for the public administration. It is made up of five themes: employee engagement; diversity and inclusion; HR management; procurement; and, technology and workplaces.

2.4.1 Employee engagement

The employee engagement theme seeks to measure the extent to which those working in the public administration feel a sense of commitment and motivation to their work and have sufficient support and development to do their jobs well. Engaged employees enable staff "to give their best and to help [their organisation] succeed – and from that flows a series of tangible benefits for [the] organisation and individual alike" (MacLeod and Clarke 2009), Borst et al. (2020) find that the relationships between work engagement and other beneficial work-related attitudes tend to be stronger for public sector employees than for those in the private sector.

2.4.2 Diversity and inclusion

The diversity and inclusion theme seeks to measure the extent to which the public administration workforce reflects the population and society it serves. Apfelbaum, Phillips, and Richeson (2014) show that workforce diversity improves the quality and impartiality of decisions, while Talmor (2022) argues when some groups are not represented or are misrepresented that anger and mistrust of institutions follows. The OECD (2015a) identifies that "a more representative public administration can better access previously overlooked knowledge, networks and perspectives for improved policy development and implementation".

2.4.3 HR management

The HR management theme seeks to measure the formal practices that govern the recruitment and management of an effective public administration workforce. As Bovaird and Loeffler (2023) note "around 70 per cent of the budgets of most public organisations are spent on staff" and thus "if the HR policies are not right, then public organisations will not attract the human resources they need to perform the functions of government and deliver the services that government has promised the electorate". Fukuyama (2013) recognises that proper recruitment and reward of public servants "remain at the core of any measure of quality of governance" as well as the importance of merit in recruitment and promotion.

2.4.4 Procurement

The procurement theme seeks to measure the operational quality of public procurement practices. The World Bank (2020) notes that "public procurement accounts on average for 13% to 20% of GDP", and that "[procurement] can play a strategic role in delivering more effective public services" (World Bank 2016).

2.4.5 Technology and workplaces

The technology and workplaces theme seeks to measure the enabling environment for public employees, the IT systems they use, the buildings they work in and the ways they work. It has long been recognised that governments need to keep up with technological advances, leveraging technology to "enhance the speed and efficiency of operations, by streamlining processes, lowering costs" (Madzova, Sajnoski, and Davcev 2013), adopting modern workplace technologies also helps officials to be more agile, creative and innovative in addition to enhancing productivity (OECD 2017). While there were moves to support remote working and leverage the opportunities of online technologies for collaboration, the COVID-19 pandemic presented a revolutionary moment for the uptake of these technologies as many governments mandated most or all of their workforces to work from home to combat the spread of the pandemic, in the post-pandemic period there are opportunities to realise continued benefits by continuing to support more diverse ways of working (OECD 2023).

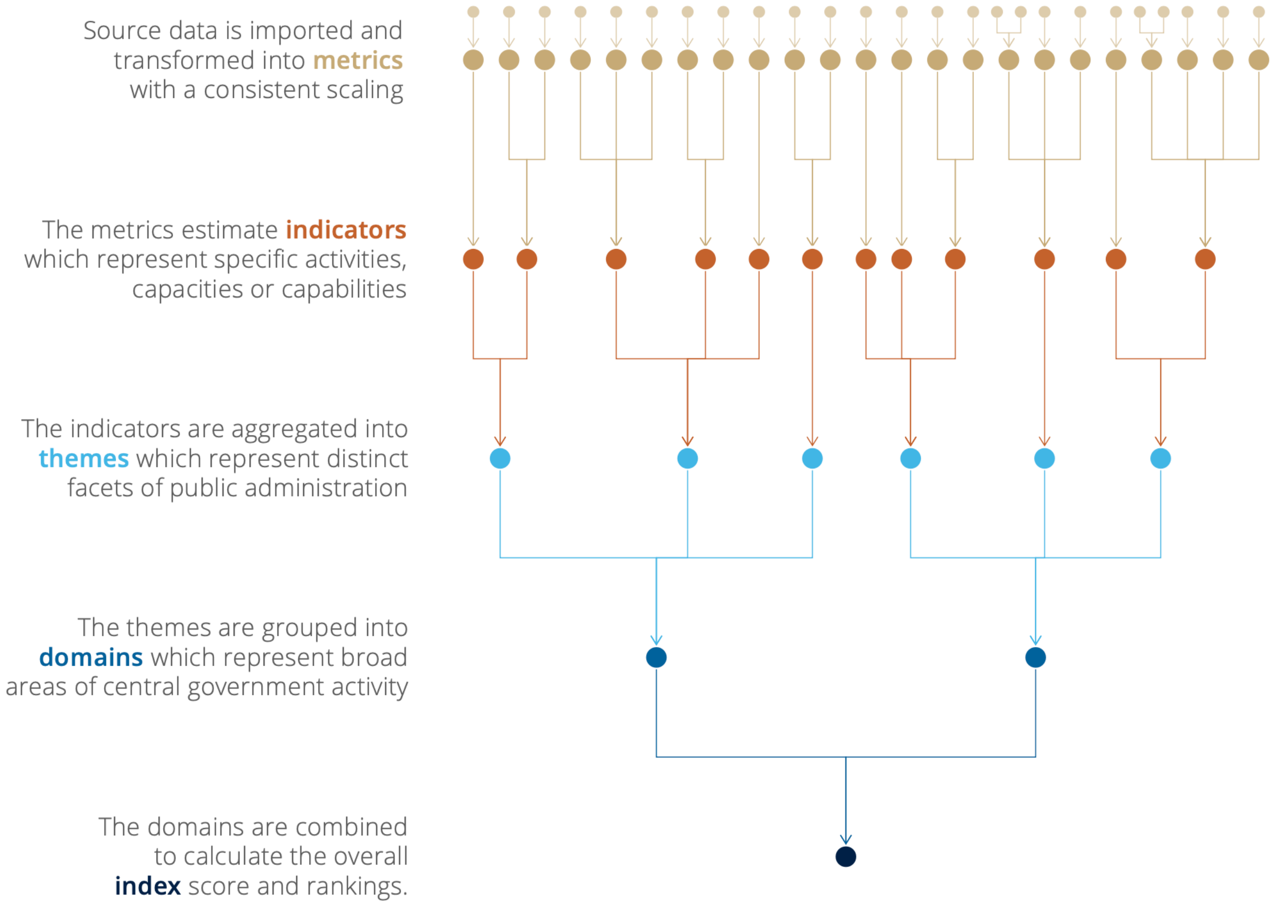
3 Data model, methodological principles and technical approach

Having defined a conceptual framework we can now define a data model that seeks to realise the conceptual framework into a measurement framework that can be used to calculate the Index and its intermediate components.

3.1 Data model

The conceptual framework outlines four domains and twenty themes, the data model expands this to define five levels. The index represents an overall comparative assessment (as far as it that possible) of a country's public administration, it is based on four domains which represent broad areas of a public administration's capacities and features. Each domain is based on five themes which represent a particular str and of public administration capacity. A theme is then based on a set of indicators which measure a specific activity or feature or quality of public administration. An indicator is estimated from on one or more metrics which are transformed and normalised versions of individual variables from source data[1]. Indicators are intended to be mutually exclusive from each other, whereas metrics may either measure further sub-divisions of the indicator or they may measure similar concepts but from different perspectives or in different ways. Figure 1 provides a visual representation of the data model.

Figure 3.1: Diagram of the Blavatnik Index's data model



[1]All metrics, except for two relating to gender equality, have a one-to-one relationship with variables in the source data. The two gender equality metrics are also based on a single variable from the source data, which variable is selected depends on data availability.

3.2 Methodological principles

Having defined the structure of the data model we can also define core principles that guide the selection of data and the development of the Index's methodology:

- The index should focus on the administrative functions of national, governments, see the sections on the Index's conceptual scope (Chapter 4) and geographic scope (Chapter 5) for more detail.
- The index should be based on data that is openly available and actionable, see the sections on selecting data sources (Chapter 6) and processing data sources (Chapter 7) for more details. There is also a section on how the metrics extracted from the source data are aligned to the conceptual framework and data model (Chapter 8).
- The index should be calculated using a simple methodology so that practitioners can easily understand how source data flows through the data model into overall scores, see the sections on country selection (Chapter 9) and data aggregation

  (1) for more details.

3.3 Technical approach and workflow

As with the InCiSE Index, the Blavatnik Index of Public Administration is based on reproducible analytical pipeline and functional programming approaches, where as far as possible all stages of the Index's data collation, processing, calculation and outputs are written in code. This approach has several benefits:

- A code-base pipeline approach makes it easier to identify and correct for errors, the entire workflow (assuming all source data has already been downloaded) of processing data sources and calculating the Index results takes less than 1 minute.
- A code-based pipeline approach can be easily re-run as and when data sources are updated, the approach taken for index calculation also makes it easy for new or different sources to be easily integrated assuming the relevant metadata files have also been changed.
- A functional programming approach makes it easier to conduct sensitivity analysis on the Index results by allowing specific aspects of the workflow to be modified while keeping others the same.
- Others can easily replicate the results by downloading the code and data sources and running the code themselves.
- Others can inspect the code used to process the data sources and calculate the Index to better understand the methodology and the choices that have been made in its construction.
- Others can modify the code for themselves, for example to run the Index on their own subset of countries, or to make different methodological choices (e.g. to exclude certain aspects/sources or to apply a weighting scheme).

The Index is operationalised through the following workflow, which is described in more detail in further sections of this report:

- The source data is downloaded from the data publisher, either as a manual download or programmatically through an API.
- The source data is processed to extract variables of interest.
- The source data is ingested and merged with metadata relating to the Index's data model.
- Country data coverage is assessed based on the normalised data and country coverage is determined.
- The source data is then transformed and re-scaled into metrics (the base level of the Index's data model), and subsequent tiers of the data model are then calculated.

The code for the Index's data processing and calculation has been written end-to-end in the R programming language, output tables, charts and graphics have been written either in R or using the Observable JavaScript libraries.

The code for the Index is stored in four Github repositories: - one to store and process the source data; - one to calculate and store the Index results; - one for further analysis; and, - one to store geographic metadata and cartographic data related to the Index.

4 Defining the Index's conceptual scope

The aim of the Blavatnik Index of Public Administration is to compare the qualities and functions of national bureaucracies – the work of officials in the achievement of a national government's objectives. While this scope sounds simple at first glance, defining the unit(s) of analysis is not, as even among countries with shared administrative cultures or histories there are idiosyncrasies in each country's own system of government.

There are three traditional approaches to defining a country's "public administration" or "civil service":

- An organisational approach which focuses on the structure of government, featuring "upstream" entities (e.g. ministries) which set the policy direction and procedural regulation for "downstream" entities (e.g. agencies) that implement policy or deliver services.
- An accounting approach which focuses on the entities which are owned/controlled by the national government and included in the "general government" category in the UN's System of National Accounts (SNA).
- An employment status approach which focuses on the body of individuals employed as civil servants.

All three of these approaches present challenges for cross-country comparisons:

- The organisational approach is influenced by the constitutional arrangements and administrative norms of a country: whether a country is a federal or unitary state, whether a country is highly centralised or decentralised in its approach to policy development and/or public service delivery, whether public services are delivered by publicly owned entities or outsourced to private sector or civil society entities.
- The accounting approach is an overly broad definition designed for the compilation of national accounts and the comparison government spending. As such, it encompasses not only national government activity but that of all public sector entities, including sub-national and municipal government, public corporations and many other entities which do tend to fall within the definitions used by practitioners and researchers when discussing the bureaucratic activities of government officials.
- Some countries have distinct employment regimes (laws/regulations) that create a separate category of employment for "civil servants" separate from other types of workers but others do not and rely on individual laws/regulations or contractual terms to define any particular additional responsibilities/privileges. Even in countries with distinct employment regimes there are variations who this regime applies to, for example whether it is limited to just national government employees or covers all public employment in the country.

It is also important to define what the Index does not intend to measure, namely policy outcomes or wider aspects of public governance:

- The Index does not measure policy outcomes (e.g. life expectancy, literacy rates, unemployment rates, incomes or economic growth). The Index seeks to provide a way of measuring and comparing bureaucracies whereas the goals for policy outcomes are set by politicians. The logic model of the Index's conceptual framework assumes that for two countries with the same goal, the country with the "better" bureaucracy ought to achieve better outcomes (all other things being equal). Even where countries have similar goals, they may take very different approaches to achieving those goals or set different targets to measure their progress.
- The Index also does not measure wider aspects of public governance or democracy such as the legislature, the judiciary or the rule of law, or media and academic freedom. These wider aspects, while crucial to understand a country's governance in full are more directly considerations for political rather than administrative actors. The Index is intended to be a practical tool to help understand how public administrations work and where countries can learn from others to drive improvements.

We use two criteria to define the scope and unit(s) of analysis for the Blavatnik Index of Public Administration:

- The Index measures national-level governments – the central/federal government entity of a country are the primary focus, but relying on "general government" or "public sector" proxies if necessary.
- The Index measures executive/administrative activities and characteristics – the activities and characteristics of the national government's executive, especially relating to the administration of government and public services.

5 Defining the Index's geographic scope

The previous International Civil Service Effectiveness (InCiSE) Index covered only OECD and EU member countries (31 countries in 2017 and 38 countries in 2019), driven somewhat by the InCiSE Index's reliance on several sources limited to either OECD or EU countries (11 of the 25 sources for the InCiSE Index 2019 had an OECD or EU member scope). In establishing the Blavatnik Index of Public Administration, we adopted a desire to produce an index with a more global scope, either in addition to or instead of an index limited to OECD/EU countries.

Following the scoping exercise to identify potential sources (see section 2) and the mapping exercise to align source data to our conceptual framework (see section 4) we were able to review geographic coverage. Of the 147 data points identified for inclusion in the Blavatnik Index of Public Administration: 70 have a "global" coverage, that is they cover a large number of countries/territories (min: 109, median: 140, max: 198), 70 are limited to OECD or EU member countries and associated partners (min: 21, median: 40, max: 51), and 7 are limited to a largely non-OECD scope (all from a single source covering 137 countries, comprising countries which were not OECD members prior to 1989).

Based on this spread of data it was decided to focus construction of the Blavatnik Index of Public Administration on the data from "global" sources. In addition to the global sources, data from the non-OECD source and its companion source with an OECD/EU scope were also selected for inclusion. As a result, the Index is made up of 82 metrics (the 70 global data points plus 7 metrics from the non-OECD source and 5 from its companion OECD source). Table 5.1 summarises the number of metrics by theme and scope. All but 4 of the 20 themes defined in the Index's conceptual framework are measured in the global Index, two themes only have data from sources with an OECD/EU scope (cross-government collaboration and employee engagement), while for one theme (social security) no relevant data was identified. While not included in the overall Index calculations, data from the OECD/EU specific sources has been retained and processed to allow for deeper analysis of particular themes.

Table 5.1: Blavatnik Index metrics by geographic coverage

	Geographic grouping			
Domain/theme	Global	OECD/EU	Non-OECD	Total included
Strategy & Leadership				
Strategic capacity	—	5	2	4
Collaboration	—	4	—	0
Openness & communications	5	10	1	7
Trust & integrity	9	5	1	10
Innovation	2	—	1	3
Public Policy				
Policy making	—	5	1	2
Financial management	—	4	—	0
Regulation	2	6	—	2
Crisis & risk management	6	—	—	6
Use of data	13	3	—	13
National Delivery				
System oversight	—	7	1	2
Digital services	3	14	—	3
Tax administration	12	—	—	12
Border services	1	—	—	1
Social security administration	—	—	—	0
People & Processes				
Employee engagement	—	1	—	0
Diversity & inclusion	10	2	—	10
HR management	5	4	—	5
Procurement	1	—	—	1
Technology & workplaces	1	—	—	1
Total number of metrics	70	70	7	82

6 Selection of data sources

The Blavatnik Index of Public Administration is based on data from 17 different sources. This section describes the approach to selecting those sources and how the source data is processed and extracted.

A scoping exercise was undertaken in early 2024 to identify potential sources for the Index, the following criteria were used to determine eligibility:

- **Open access** – the data source and its methodology must be published online in a free-to-access form.
- **Actionable** – the data must measure some quality or component of public administration that officials and/or ministers can act on to improve performance.
- **Quantifiable** – if not already directly numeric, the data must be easily quantifiable.
- **Recency** – the source should have been updated since 2019.

6.1 Selection criteria

The ideal for the open access criterion is a data source that publishes not only its high-level results but also the raw scores for a country and its methodology. Many but not all sources used in the Blavatnik Index meet this requirement, some sources do not publish raw scores for countries but do publish an overall score and sometimes intermediate scores but alongside this publish a full methodology that allows for external scrutiny of how countries are assessed.

The open access criterion is designed to support the second criterion, that the data must be actionable. Whether data is actionable is considered in two ways: conceptually and practically. Data is conceptually actionable if it measures something that we can expect government officials or ministers to respond to. Data is practically actionable if government officials or ministers can determine how their country scores and respond accordingly.

The Blavatnik Index of Public Administration is by its nature a quantitative exercise, data must therefore either be already quantitative in nature or be measured in such a way that it is easy (and sensible) to coerce into a quantitative form, for example the conversion of an ordinal scale into a numeric range or summing related binary (yes/no) categories.

Only a few of the data sources used by the Blavatnik Index of Public Administration are updated annually, some are updated biennially and others on slower timescales. While the Blavatnik Index of Public Administration is not directly comparable to the previous InCiSE Index, it was decided to only include data sources that had been updated or published after the publication of the 2019 edition of the InCiSE Index.

6.2 Data source discovery and scoping exercise

The scoping exercise considered 36 potential sources and 18 are used in the calculation of the Blavatnik Index of Public Administration 2024. Table 3 1 provides a list of the sources used, and considered but not used, in the Blavatnik Index of Public Administration.

Initially, the scoping exercise identified these sources for consideration by reviewing the status of sources used for the InCiSE Index's 2017 and 2019 editions as well as reviewing other datasets and outputs published by the same institutions since the 2019 report. The exercise also reviewed the European Joint Research Centre's Composite Indicators and Scorecards Explorer that might be of relevance to public administration and reviewing both these products themselves and their sources.

When considering indexes and composite indicators which combine data from other sources the scoping exercise would to identify the original sources. For example, reviewing sources published by the World Bank identified the Statistical Performance Indicators, further review of this product's sources identified Open Data Watch's Open Data Inventory and the OECD's PARIS21 indicators as original sources for review.

In the initial scoping exercise sources were considered if they had either a global focus or, like InCiSE 2019, were limited to OECD/EU members. Following the scoping exercise, it was determined that there was sufficient data to enable a global index to be produced.

Table 6.1: Sources reviewed for the Blavatnik Index oof Public Administration 2024

Source grouping	Data sources
Sources used in the Blavatnik Index of Public Administration 2024 and in the InCiSE Index 2019	• Doing Business Report (World Bank) • Gender Statistics Database (European Institute for Gender Equality) • Global Corruption Barometer (Transparency International) • Global Data Barometer (D4D.net and ILDA) • ILOSTAT (International Labor Organization) • International Survey of Revenue Administration (CIAT, IMF, IOTA, and OECD) • Open Budget Survey (International Budget Partnership) • Quality of Government Expert Survey (University of Gothenburg) • Rule of Law Index (World Justice Project) • Sendai Framework Monitor (United Nations) • Sustainable Governance Indicators (Bertelsmann Stiftung) • Varieties of Democracy dataset (University of Gothenburg)
Sources used in the Blavatnik Index of Public Administration 2024 but not in the InCiSE Index 2019	• Bertelsmann Transformation Index (Bertelsmann Stiftung) • Global Cybersecurity Index (International Telecommunications Union) • GovTech Maturity Index (World Bank) • Logistics Performance Index (World Bank) • Open Data Inventory (Open Data Watch) • PARIS21 Statistical Capacity Monitor (OECD)
Source covering only OECD/EU countries and not included in the the Blavatnik Index of Public Administration 2024	• Digital Government Index (OECD) • eGovernment Benchmark Report (European Commission) • Indicators of Regulatory Policy and Governance (OECD) • OUR Data Index (OECD) • Public Integrity Indicators (OECD) • Public Service Leadership and Capability Survey (OECD) • OpenTender and Global Public Procurement Dataset (Government Transparency Institute)
Sources with global coverage but not included in the Blavatnik Index of Public Administration 2024	• Corruption Perceptions Index (Transparency International) • E-Government Development Index (United Nations) • Global Innovation Index (World Intellectual Property Organisation) • Global Survey of Public Servants • OECD Budgeting Databases (OECD) • Open Budget Survey (International Budget Partnership) • Worldwide Governance Indicators (World Bank)
Sources used in the InCiSE Index 2019 but not updated	• Dataset on Public Procurement (OECD) • FMIS and OBD dataset (World Bank) • Global Competitiveness Report Executive Opinion Survey (World Economic Forum) • Global Open Data Index (Open Knowledge Foundation) • Survey of Adult Skills/Programme for the International Assessment of Adult Competencies (PIAAC) (OECD) • Survey on the Governance of Critical Risks (OECD)

6.3 Sources used in InCiSE 2019 and the Blavatnik Index

There are 12 sources which were used in the InCiSE 2019 Index and which have been selected for inclusion in the Blavatnik Index of Public Administration 2024:

- The World Bank's Doing Business report collected a range of measures relating to the ability to start and continue a business in 190 different countries and territories. One of these measures is an estimate of the time in a given year it takes businesses to file and pay taxes. This measure was used by InCiSE 2019 and has been retained. The Doing Business report was discontinued in 2020 and is being replaced by a new product called Business Ready but this was not available at the time of scoping and production of the Blavatnik Index of Public Administration.
- The European Institute for Gender Equality's Gender Statistics Database is used by the OECD as the source for an indicator in its Government at a Glance publication on the representation of women in senior levels of government, which was used by InCiSE 2019. For the Blavatnik Index of Public Administration we have sourced data directly from the EIGE's database.
- Transparency International's Global Corruption Barometer is a public opinion survey that collects data on public perceptions and experiences of corruption and bribery. One measure within the survey is the perception of how many government officials are involved in corruption, this was used by InCiSE 2019 and has been retained.
- The Global Data Barometer is published by the Data for Development Network and ILDA, that measures the range and nature of "data for public good" across topics including climate, company information, health, land, political integrity, public finance and public procurement, it also seeks to measure data governance and data capability. The Barometer is a successor to World Wide Web Foundation's Open Data Barometer which was used in InCiSE 2019.
- The International Labor Organization's ILOSTAT service aggregates official statistics from around the world on labour market matters, includes public sector employment. Measures on the public sector employment by sex from ILOSTAT were used in InCiSE 2019 and have been retained.
- The International Survey of Revenue Administration is a joint exercise of the Inter-American Center of Tax Administrators, the International Monetary Fund, the Intra-European Organisation of Tax Administrators and the OECD. The ISORA is used by the OECD to compile its Tax Administration Series, which was used by InCiSE 2019.
- The University of Gothenburg's Quality of Government Expert Survey was used by InCiSE 2019 as a source of measures relating to HR management and integrity, some of these measures are now included within the dataset produced by the University of Gothenburg's Varieties of Democracy project.
- The World Justice Project's Rule of Law Index is produced from a mix of public and expert perception surveys about the rule of law and the operation of justice including measures relating to administrative procedures and public integrity. This was used by InCiSE 2019 and has been retained.
- The United Nation's Sendai Framework Monitor monitors countries adoption and adherence to the Sendai Framework for Action on natural disaster risk reduction. The Sendai Framework is the successor to the Hyogo Framework for Action, monitor data for which was included in InCiSE 2019.
- The Bertelsmann Stiftung's Sustainable Governance Indicators measure a number of facets of governance including policy outcomes, the quality of democracy, and good governance. Several of its measures relating to the government executives and public administration were used by InCiSE 2019, some of which have been retained and matched with similar measures in the Bertelsmann Transformation Index.

6.4 Sources used by the Blavatnik Index but not used by InCiSE 2019

There are 6 sources which were not used by the InCiSE 2019 Index but which have been identified for inclusion in the Blavatnik Index of Public Administration 2024:

- The Bertelsmann Stiftung's Bertelsmann Transformation Index is similar to the Sustainable Governance Indicators which were used by InCiSE 2019 but covers non-OECD/EU countries and so was not included in InCiSE 2019. Measures relating to government executives and public administration have been included in the 2024 Index.
- The International Telecommunication's Global Cybersecurity Index measures country's developments in cybersecurity defences and capability across five pillars (legal measures, technical measures, organizational measures, capacity development and cooperation).
- The World Bank's GovTech Maturity Index measures countries' efforts in the digital transformation of the public sector, it assesses core policies and technologies, the provision of digital public services, the use of digital methods for citizen engagement, and policies and practices to support digital innovation and transformation.
- The World Bank's Logistics Performance Index includes a survey of international logistics operators on their experiences of trade and shipping, including measures relating to the efficiency and ease of completing customs obligations.
- Open Data Watch's Open Data Inventory is a biennial assessment of the availability and coverage of official statistics across the domains of economic, environmental and social statistics.
- The Statistical Capacity Monitor produced by PARIS21 (a collaboration of the UN, European Commission, OECD, IMF and World Bank; hosted by the OECD secretariat) assesses countries' capacities in regards official statistics, it collates data from other indicators, including the Open Data Inventory (see above), as well as some primary

data collection. The Statistical Capacity Monitor's measure on the use of statistics in national policy documents has been included in the 2024 Index.

6.5 Sources limited to OECD/EU member countries

There are 8 sources limited to OECD/EU members that were either used by InCiSE 2019 or are new and considered suitable for inclusion in a 2024 Index if it were limited to OECD/EU countries:

- The OECD's Digital Government Index provides an overview of government policies and practices in digital government across six aspects: data-driven public sector, digital by design, government as a platform, open by default proactiveness, and user-driven. The Digital Government Index was first published after InCiSE 2019.
- The European Commission's eGovernment Benchmark Report is a comprehensive assessment of over 5,000 public sector websites in EU member and neighbouring countries, it was used as a data source for InCiSE 2019.
- The OECD's Indicators of Regulatory Policy and Governance assess government's policies and practices in the development of legislation and regulations, it was used as a data source for InCiSE 2019.
- The OECD's OUR Data Index assesses government's policies and practices in the publishing of open data, it was used as a data source for InCiSE 2019.
- The OECD's Public Integrity Indicators provide a range of public integrity and anti-corruption measures including: information disclosure; open government practices; conflict of interest, lobbyist transparency and regulation, senior officials' post-employment activities; and financial control, risk management and internal audit. The Public Integrity Indicators were first published after InCiSE 2019.
- Measures collected via OECD's Public Service Leadership and Capability Survey have been published in its Government at a Glance report, including measures relating to learning and development, gender diversity practices, employment and recruitment frameworks, mobility and flexible working arrangements. The Public Service Leadership and Capability Survey was first collected and published after InCiSE 2020.
- OpenTender is a project of the Government Transparency Institute first developed under the EU's Horizon 2020 research fund to provide analysis of public procurement tender notices and contract data across EU members, with a focus on integrity and anti-corruption. In April 2023, the Government Transparency Institute published an expanded dataset (the Global Public Procurement Dataset) based in part on the data from the OpenTender project but expanded to also include data from Armenia, Bangladesh, Brazil, Colombia, Georgia, Kenya, Indonesia, Paraguay, Uruguay and the United States.

6.6 Sources considered but not included

There are 6 sources which were identified during the scoping exercise but following further review it was decided not to include:

- Transparency International's Corruption Perceptions Index (CPI) is a composite index drawing on a large number of sources, some of which had already identified for use in the Blavatnik Index of Public Administration, as a result the CPI is not used. However, data from Transparency International's Global Corruption Barometer which measures public opinions of corruption has been selected for inclusion.
- The UN's E-Government Development Index (EGDI) is regularly used as a benchmark for international comparisons of governments' progress in digital government, however it fails to meet our eligibility for inclusion. While high-level results are published for the EGDI overall and its sub-components, the underlying questionnaire that is used to calculate the Index and/or country scores to those questions are not. It is therefore not possible to independently determine the criteria used to assess a country or to identify how a country can improve its score in the EGDI.
- The World Intellectual Property Organization's Global Innovation Index (GII) measures the innovative capacity of countries and economies. Innovation was identified in the conceptual framework for the InCiSE 2017 and 2019 Indexes but was not measured, it is also included in the conceptual framework for the Blavatnik Index of Public Administration, therefore the Global Innovation Index was reviewed as a potential source for metrics. The majority of the Global Innovation Index's metrics relate to private sector activity, its public sector metrics were either: (i) measure something out of scope such as policy goals (e.g. science graduates); (ii) already identified for use in the Blavatnik Index of Public Administration (e.g. the Logistics Performance Index); (iii) a source deemed out of scope (e.g. the World Bank's Worldwide Governance Indicators, see below); or (iv) a source that has been rejected (e.g. the UN's E-Government Development Index).
- The Global Survey of Public Servants (GSPS), a collaboration of Stamford University, University College London, University of Nottingham and the World Bank, is a dataset that collates survey results from employee attitudes surveys of officials working in central government organisations. The dataset provides data from surveys conducted in 28 countries with some results harmonised to one of 25 common concepts. Unlike other datasets considered in the scoping exercise the country coverage does follow regional/economic patterns (at least two

countries are included from each continent), coverage also varies considerably by topic and harmonisation concept. Once filtering the results for data that only relates to 2019 or later, no harmonised concept has data for more than 9 countries.

- The OECD has collected a number of datasets relating to budgeting and financial management practices, for example on gender budgeting, green budgeting, spending reviews, these were reviewed for potential indicators. While these datasets provide useful comparative information on budgeting and financial management practices it was not deemed feasible to coerce this information into numerical measures that could discern differing levels of 'performance'.
- The World Bank's Worldwide Governance Indicators (WGI) are a set of aggregate indicators measuring six topics: voice and accountability; political stability and absence of violence/terrorism; government effectiveness; regulatory quality; rule of law; and, control of corruption. The WGIs make use of a large array of sources, including many already identified for inclusion in the Blavatnik Index of Public Administration. In addition to this potential for "double counting", there are two further limitations to the WGIs:

    1. The source data for the WGIs is input to a statistical model to account for missing data and to calculate the six aggregate scores, while the methodology and weightings of this model are published it is not simple for countries to determine how their scores in the source data relate to their overall score and where to focus actions for improvement.
    2. Several of the sources are closed or proprietary in nature, while the World Bank publishes overall country scores from these sources the underlying methodology and/or "raw" scores are not readily available, therefore making it difficult for countries to determine where to focus actions for improvement.

6.7 Sources used in InCiSE 2019 but not updated

There are sources which were used in InCiSE 2019 which have not been updated since that report was published:

- The OECD's Dataset on Public Procurement assessed the public procurement policies and practices of central governments in OECD member countries.
- The World Bank's FMIS and OBD dataset provided information on the existence of financial management information systems and the range of financial management data that is published.
- The World Economic Forum's Global Competitiveness Report Executive Opinion Survey was used for its measure on business opinions on the efficiency of public spending. Since InCiSE 2019 the Global Competitiveness Report has been discontinued, while an Executive Opinion Survey is to support the compilation of the WEF's Global Risks Report this does not cover the same information as that used for the Global Competitiveness Report and is not published in a similar format.
- The Open Knowledge Foundation's Global Open Data Index was used as one of several measures relating to open data.
- The OECD's Survey on the Governance of Critical Risks was used to provide metrics for InCiSE 2019's measures on crisis and risk management, this dataset has not been updated.
- The OECD's Survey of Adult Skills (Programme for the International Assessment of Adult Competencies, PIAAC) provides measures of literacy, numeracy and other skill levels and was used to provide metrics for InCiSE 2019's measures relating to staff capabilities. The experimental design for PIAAC envisages it being repeated on a decadal basis, the second round of data collection has been completed, but the results and data were yet published at the time of data source scoping and production of the Index.

7 Processing of data sources

Having identified and selected sources for inclusion in the Index, each source then underwent some degree of processing to ensure all source data is ultimately held in the same, common format.

The common format is defined as a comma-separated values (CSV) file where each row represents a single data point (i.e. each row relates to a specific country's value for a specific variable for a specific year). The common format has four columns:

- *cc_iso3c* – a 3-character alphabetic code identifying the country/territory, based on the ISO 3166-1 standard.
- *ref_year* – a 4-digit number providing the reference year of the data point.
- *variable* – a variable label that provides a unique reference to the data series the data point is drawn from.
- *value* – the numeric value of the data point.

The processing of data sources also involved the creation of a metadata record for each source that provided standardised metadata about the source, a description of the source, documentation of the processing, and metadata relating to the extracted variables.

For many sources this processing was a simple process that involves the extraction of the relevant subset of variables, conformance of country identifiers to the Index's harmonised coding scheme, and transposition of the data from a 'wide' to 'long' format.

In some situations, more complex processing is undertaken such as when data for countries in different regions has been published in separate files and needs consolidation, where data has been published in PDF files or to in situations where it has been necessary to re-process the original data to extract relevant scores. See the section on pre-processing (Appendix C) for more details.

The source code for the processing is available on Github.

Table 7.1: Type of processing applied to data sources

Type of processing	Data sources
Variable extraction and formatting	• Bertelsmann Transformation Index (Bertelsmann Stiftung) • Doing Business Report (World Bank) • Gender Statistics Database (European Institute for Gender Equality) • ILOSTAT (International Labor Organization) • Logistics Performance Index (World Bank) • PARIS21 Statistical Capacity Monitor (OECD) • Quality of Government Expert Survey (University of Gothenburg) • Rule of Law Index 2023 (World Justice Project) • Sendai Framework Monitor (United Nations) • Sustainable Governance Indicators (Bertelsmann Stiftung) • Varieties of Democracy dataset (University of Gothenburg)
Consolidation of regional data files	• Global Corruption Barometer (Transparency International)
Extraction of data from PDF report	• Global Cybersecurity Index (International Telecommunications Union)
Re-processing of original data	• Global Data Barometer (D4D.net and ILDA) • GovTech Maturity Index 2022 (World Bank) • International Survey of Revenue Administration (CIAT, IMF, IOTA, and OECD) • Open Data Inventory (Open Data Watch)

8 Alignment of sources to the Index's conceptual framework

In order to calculate the Index, and its components, the source data needs to be aligned against our conceptual framework, in. doing so the following approach was followed:

- To ensure simplicity of the methodology for calculating the Index, variables from the source data were allocated to only one theme.
- When selecting the theme to allocate a variable to, maximising coverage of the conceptual framework has been prioritised.
- Variables measuring similar concepts are grouped together to define the indicators within that theme.

This section describes the relationship of source data to the Index's conceptual framework and data model. For each theme, a table details the indicators and variables that make up the theme, it also lists any transformation of the variables and metadata about the variable/source (reference year(s), number of countries covered, type of data collection).

To enable effective comparison, all source data is translated to a 0-to-1 scale, in addition to scaling some data undergoes a transformation (e.g. data where higher scores relate to poorer performance are inverted) before it is scaled, Appendix D describes the different transformation processes outlined in the metrics tables.

The type of data collection used for each of the sources have been coded into one of nine types, which can be grouped into three broad groupings (quantitative data, assessments, and opinion measures):

- Quantitative data
  - Administrative data – published data drawn from administrative systems (e.g. data submitted by tax agencies to the International Survey of Revenue Administration).
  - Official statistics – published data drawn from official statistics (e.g. official statistics data collated by the International Labor Organization).
  - Statistical analysis – statistical analysis conducted on government data/publications (e.g. the analysis of policy documents used by the PARIS21 Statistical Capability Monitor).
- Assessments
  - Expert assessment – external objective assessment of government practices (e.g. assessment and analysis of open data published by governments by civil society researchers for the Global Data Barometer).
  - Government self-assessment – responses by government officials to a survey (e.g. responses by officials to the World Bank survey that is used to calculate the GovTech Maturity Index).
- Opinion measures
  - Business opinion – responses by businesses to an opinion survey (e.g. responses by businesses to the World Bank survey used to collect information for the Logistics Performance Index).
  - Expert opinion – responses by experts to an opinion survey (e.g. responses by academics/civil society experts to the Quality of Government Expert Survey), or assessments by experts that provide a rating (e.g. academic/civil society expert ratings used to produce in the Bertelsmann Transformation Index).
  - General public opinion – responses by the general public to an opinion survey (e.g. responses by members of the public to the survey used to calculate the Global Corruption Barometer).
  - Combined expert and public option – the combination of responses from surveys of experts and the general public (e.g. the combination of a survey of legal professionals and a general public opinion survey used in the production of the Rule of Law Index).

Table 8.1: Data sources used in the Blavatnik Index of Public Administration 2024

Data source	Publisher	Type of data collection	Reference year(s)	Website
Bertelsmann Transformation Index 2024 (BTI)	Bertelsmann Stiftung	Opinion (expert)	2023	https://bti-project.org
Doing Business 2020 (DB)	World Bank	Opinion (business)	2020	https://www.worldbank.org/en/businessready/doing-business-legacy
Gender Statistics Database (GSD)	European Institute for Gender Equality	Statistical data (official statistics)	2023	https://eige.europa.eu/gender-statistics/dgs
Global Corruption Barometer (GCB)	Transparency International	Opinion (general public)	2019-2021	https://www.transparency.org/en/gcb
Global Cybersecurity Index 2020 (GCI)	International Telecommunications Union	Assessment (self-assessment)	2020	https://www.itu.int/epublications/publication/D-STR-GCI.01-2021-HTM-E
Global Data Barometer (GDB)	Data for Development Network	Assessment (external)	2021	https://globaldatabarometer.org
GovTech Maturity Index 2022 (GTMI)	World Bank	Assessment (self-assessment)	2022	https://www.worldbank.org/en/programs/govtech/gtmi
ILOSTAT (ILO)	International Labor Organisation	Statistical data (official statistics)	2023	https://ilostat.ilo.org
International Survey of Revenue Administration 2021 (ISORA)	CIAT, IMF, IOTA and OECD	Statistical data (administrative data return)	2021	https://data.rafit.org
Logistics Performance Index (LPI)	World Bank	Opinion (business)	2022	https://lpi.worldbank.org
Open Data Inventory 2022-23 (ODIN)	Open Data Watch	Assessment (external)	2022	https://odin.opendatawatch.com
Quality of Government Expert Survey 2020 (QOG)	Quality of Government Institute, University of Gothenburg	Opinion (expert)	2020	http://qog.pol.gu.se
Rule of Law Index 2023 (ROLI)	World Justice Project	Opinion (expert & general public)	2016-2022	https://worldjusticeproject.org/rule-of-law-index
Sendai Framework Monitor (SFM)	UN	Assessment (self-assessment)	2017-2023	https://unstats.un.org/sdgs/dataportal
Statistical Capacity Monitor (SCM)	PARIS21 (UN, EC, OECD, IMF & WB)	Assessment (expert)	2019-2020	https://statisticalcapacitymonitor.org
Sustainable Governance Indicators 2022 (SGI)	Bertelsmann Stiftung	Opinion (expert)	2022	http://sgi-network.org
Varieties of Democracy dataset version 14 (VDEM)	V-Dem Institute, University of Gothenburg	Opinion (expert)	2023	https://www.v-dem.net

8.1 Strategy and Leadership domain

The strategy and leadership domain has five themes: strategic capacity; cross-government collaboration; openness and communications; integrity; and, innovation. Of these, all but cross-government collaboration can be measured from sources with global coverage.

8.1.1 Strategic capacity

There are two sources for the strategic capacity theme: the Bertelsmann Transformation Index (BTI) and the Sustainable Governance Indicators (SGI). Four measures are extracted from these sources and used to estimate one indicator: prioritisation.

The prioritisation indicator measures the extent to which governments can set priorities/strategies. It is estimated by two measures from the SGI measuring the influence of strategic planning units and the strategic appraisal capability of the centre of government, these are complemented by the BTI measures on the setting of strategic priorities and the efficiency of resource allocation.

A further three measures were identified from OECD/EU only sources, these could be used to estimate a second indicator on institutional adaptation.

- Metrics table: Table 8.2
- Framework definition: Section 2.1.1

8.1.2 Cross-government collaboration

Four measures from the Sustainable Governance Indicators were identified as potential metrics for the collaboration theme, which could be used to estimate two indicators: collaboration in policy development, and international collaboration; however, there were no appropriate measures in the Bertelsmann Transformation Index that these could be matched with. Therefore, the collaboration theme is not measured in the global version of the index.

- Metrics table: *not available*
- Framework definition: Section 2.1.2

8.1.3 Openness and communications

There are four sources for the openness and communications theme: the Bertelsmann Transformation Index (BTI), the GovTech Maturity Index, the Rule of Law Index, and the Sustainable Governance Indicators (SGI). Seven measures are extracted from these sources and are used to estimate three indicators: right to information; open government; and, engagement and feedback.

The right to information indicator assesses the extent to which non-government actors/members of the public can obtain/request access to official information. It is estimated from the ROLI measure on right to information.

The open government indicator assesses the extent to which non-government actors/members of the public can access laws/regulations and the extent to which governments make information proactively available to the public. It is estimated from the ROLI measure on publicized laws and government data and a measure extracted from the GTMI on the existence of portals for open government initiatives and open data.

The engagement and feedback indicator assesses the extent and nature of government consultation in the development of policies and regulations and whether service users can provide feedback about their experience of public services. It is estimated from a measure from the SGI matched with a measure from the BTI, both measuring consultation with non-state actors, the ROLI measure on complaint mechanisms and a measure extracted from the GTMI on digital methods for public participation.

A further 9 measures were identified from OECD/EU only sources, these would add further depth to the three global indicators as well as estimate a fourth indicator around public communications.

- Metrics table: Table 8.3
- Framework definition: Section 2.1.3

8.1.4 Integrity

There are six sources for the integrity theme: the Bertelsmann Transformation Index (BTI), the Global Corruption Barometer (GCB), the Global Data Barometer, the Quality of Government Expert Survey (QOG), the Rule of Law Index (ROLI), and the Varieties of Democracy Dataset (VDEM). 10 measures are extracted from these sources and are used to estimate four indicators: impartial behaviour; corruption; sanctions; and, integrity data.

The impartial behaviour indicator measures the degree of impartiality in the operation of public administrations, i.e. the extent to which politicians interfere with day-to-day administrative practice and delivery of public services. It is estimated by the QOG measures on whether officials act with impartiality and political interference in bureaucratic decision making, the ROLI measure on whether due process is respected in administrative proceedings, and the VDEM measure on whether public officials are impartial in the performance of their duties.

The corruption indicator measures the extent to which officials are not involved in corruption/corrupt practices. It is estimated by the GCB measure on public perception of the extent of officials involved in corruption, and the VDEM measures on whether public officials grant favours in return for a bribe and whether officials misuse or embezzle public funds/resources.

The sanctions indicator measures the extent to which officials found to engage in misconduct are sanctioned for their misbehaviour. It is estimated by the BTI measure on the extent to which officials are prosecuted and the ROLI measure on the extent to which officials are sanctioned.

The integrity data indicator measures the availability of data published by government relating to public integrity (e.g. registers of interests, gifts, meetings etc). It is estimated by the GDB's measure on integrity data availability.

A further five metrics were identified in OECD/EU only sources, these would provide estimates for two further indicators: the existence of an integrity framework; and, practices for managing conflicts of interest.

- Metrics table: Table 8.4
- Framework definition: Section 2.1.4

8.1.5 Innovation

There are three sources for the innovation theme: the Bertelsmann Transformation Index (BTI), the GovTech Maturity Index (GTMI), and the International Survey of Revenue Administration (ISORA). Three measures are extracted from these sources to measure one indicator: innovative approach.

The innovative approach indicator measures the extent to which governments are adopting or supporting innovative practices in the development of policy or delivery of public services. It is estimated by the BTI measure rating how innovative and flexible governments are overall, data extracted from the GTMI relating to the support for innovation (with a focus on digital services/new technology), and the use of innovative practices by the tax administration.

- Metrics table: Table 8.5
- Framework definition: Section 2.1.5

8.2 Public Policy domain

The public policy domain has five themes: policy making; financial management; regulation; crisis and risk management; and, use of data and statistics. All five themes in the public policy domain can be measured from global sources.

8.2.1 Policymaking

There are two sources for the policymaking theme: the Bertelsmann Transformation Index (BTI) and the Sustainable Governance Indicators (SGI). Two measures are extracted from these sources to estimate one indicator: policy coordination.

The policy coordination indicator measures the extent to which government design coordinated policies. It is estimated by the SGI measure on the ministerial/cabinet coordination matched with the BTI measure on coordinating conflicting objectives.

A further four measures were identified in OECD/EU only sources, these would provide estimates of two additional indicators: policy expertise; and, policy evaluation.

- Metrics table: Table 8.6
- Framework definition: Section 2.2.1

8.2.2 Financial management

Three measures were identified from the Open Budget Survey relating to the transparency of budget documents, budget consultation practices and the transparency of budget oversight, as well as one measure from the Global Data Barometer relating to the availability of public finance data. However, on further review it was decided not to include these measures owing to the potential for misinterpretation of the results. These measures focus principally on the transparency of fiscal information provided by the government, while this is an important component it does not measure important internal practices such as the use of spending reviews, cost benefit analysis, spend controls and authorisation processes, internal record keeping/management information or internal audit processes. Moreover, several of the countries that score well in these transparency measures have well documented instances of corruption, and/or make use of national security legislation or extrabudgetary measures to hide some degree of their public expenditure.

More traditional measures of public financial management use output/outcome indicators such as the debt to GDP ratio, tax revenue as a share of GDP, public expenditure as a share of GDP, yields on government bonds. While good practices in the management of the public finances is likely an important factor in these indicators, they are much more likely to be influenced by wider economic factors and specific political decisions, meaning that these types of measures are out of scope.

In keeping with our principle for a simple and transparency methodology, since we have been unable to identify a simple approach to accounting for either corruption or the hiding of public expenditure it was decided not to include measures on financial management in the calculation of the Index at this stage.

- Metrics table: *not available*
- Framework definition: Section 2.2.2

8.2.3 Regulation

There is one source for the regulation theme: the Rule of Law Index (ROLI). Two measures are extracted from this source to estimate a single indicator: regulatory enforcement.

The regulatory enforcement indicator measures the extent to which regulations are enforced effectively and properly. It is estimated from the ROLI measures on the effective enforcement of regulations and the enforcement of regulation without improper influence.

A further six measures were identified in OECD/EU only sources, one measure would add to the existing metrics estimating the regulatory enforcement indicator while the others would estimate an additional indicator on the use of regulatory impact assessment.

- Metrics table: Table 8.7
- Framework definition: Section 2.2.3

8.2.4 Crisis and risk management

There are two sources for the crisis and risk management theme: the Global Cybersecurity Index (GCI) and the Sendai Framework Monitor (SFM). Six measures are extracted from these sources to estimate two indicators: natural hazards; and, cybersecurity.

The natural hazards indicator measures the extent to which governments have strategies to respond to natural hazard disasters. It is estimated from the SFM measure on disaster risk reduction strategies.

The cybersecurity indicator measures the extent to which governments have strategies and practices in places to handle cybersecurity risks. It is estimated from the five pillars of the GCI: capacity building measures, cooperative measures, legal measures, organisational measures, and technical measures.

- Metrics table: Table 8.8
- Framework definition: Section 2.2.4

8.2.5 Use of data and statistics

There are three sources for the use of data and statistics theme: the Global Data Barometer (GDB), the Open Data Inventory (ODIN) and the PARIS21 Statistical Capacity Monitor (PARIS21). 13 measures are extracted from these sources to estimate four indicators: data capability; data availability; data coverage; and data re-use.

The data capability indicator measures the extent of government support for data skills and capabilities, and governance policies for data and statistics. It is estimated from the GDB measures on data governance and data capability.

The data availability indicator measures the availability of official statistics and other open data published by government. It is estimated from the GDB measures on the availability of company data, land data, health data and climate data, and the ODIN measures on the openness of economic statistics, social statistics and environmental statistics.

The data coverage indicator measures the extent of indicators and disaggregation published and availability of time-series data. It is estimated from the ODIN measures on the coverage of economic, social and environmental statistics.

The data re-use indicator measures the extent to which data and statistics are used/re-used. It is estimated from the PARIS21 measure on the use of statistics in policy documents.

A further three measures were identified in OECD/EU only sources, these would provide further metrics for the data availability and data re-use indicators.

- Metrics table: Table 8.9
- Framework definition: Section 2.2.5

8.3 National Delivery domain

The national delivery domain has five themes: system oversight, digital services, tax administration, border services, and social security. Of these, all but social security can be measured from global sources.

8.3.1 System oversight

There are two sources for the system oversight theme: the Bertelsmann Transformation Index (BTI) and the Sustainable Governance Indicators (SGI). Two measures are extracted from these sources to estimate one indicator: policy implementation.

The policy implementation indicator measures the extent to which governments are able to achieve objectives. It is estimated by the SGI measure on the extent to which government can achieve its policy objectives matched with the BTI measure on whether the government can implement its policies.

A further six measures were identified in OECD/EU only sources, these would provide estimates for two additional indicators: policy monitoring and subnational relations.

- Metrics table: Table 8.10
- Framework definition: Section 2.3.1

8.3.2 Digital services

There is one source for the digital service theme: the GovTech Maturity Index (GTMI). Three measures are extracted from this source, each estimating one indicator: digital strategy and policies; infrastructure; and, digital public services.

The digital strategy and policy indicator measures the existence of strategies, policies and practices that support the government's approach to digital services. It is estimated from data extracted from the GTMI on digital strategies and policies.

The infrastructure indicator measures the existence of technologies that support the delivery of digital services. It is estimated from data extracted from the GTMI on back-end technologies and infrastructure.

The digital public services indicator measures the extent of digital services for end-users. It is estimated from the data extracted from the GTMI on the existence of service portals, e-payment services and use of digital signature technologies.

A further 14 measures were identified in OECD/EU only sources, these would provide further metrics to estimate the digital strategy and policy, the infrastructure, and the digital public services indicators, it would also provide metrics to estimate an additional indicator on user-centric service design and development.

- Metrics table: Table 8.11
- Framework definition: Section 2.3.2

8.3.3 Tax administration

There are two sources for the tax administration theme: the Doing Business report (DB), and the International Survey of Revenue Administration (ISORA). 12 metrics are extracted from these sources to estimate three indicators: tax agency administration; tax compliance; and, digital tax services.

The tax agency administration indicator measures the administrative functioning of the tax agency. It is estimated by the ISORA measures on administrative costs (as a percent of net revenue) and the turnover of tax agency staff.

The tax compliance indicator measures the compliance of taxpayers with their obligations and the effectiveness of the tax agency in enforcing compliance. It is estimated by the ISORA measures on the rate of taxes filed on time (for both corporate and personal taxes), the rate of taxes paid on time (for both corporate and personal taxes), and tax debt (as a percent of net revenue), it is also estimated by the DB measure on the annual time businesses must take to file tax returns/declarations.

The digital tax services indicator measures the use of digital methods for interacting with the tax agency. It is estimated from the ISORA measures on the proportion of taxes filed electronically (for both corporate and personal taxes), the proportion of taxes paid electronically, and the proportion of service contacts that occur via digital channels.

- Metrics table: Table 8.12
- Framework definition: Section 2.3.3

8.3.4 Border services

There is one source for the border services theme: the Logistics Performance Index (LPI). One measure is extracted from this source to estimate a single indicator: customs processes.

The customs processes indicator measures the ease with which goods can be shipped across international borders. It is estimated by the LPI measure on the efficiency of customs processes.

- Metrics table: Table 8.13
- Framework definition: Section 2.3.4

8.3.5 Social security

The GovTech Maturity Index provides data on the existence of portals for social security insurance and job searches. These are included in the measurement of the digital public services indicator in the digital services theme. No other measures of the functioning and practice of national-level social security services have been identified.

- Metrics table: *not available*
- Framework definition: Section 2.3.5

8.4 People and Processes domain

The people and processes domain has five themes: employee engagement; diversity & inclusion; HR management; procurement; and, technology and workplaces. Of these, only diversity and inclusion, HR management and procurement can be measured from global sources.

8.4.1 Employee engagement

Measures relating to learning and development practices were identified in one OECD/EU only source. There were no measures identified with extensive country coverage that would allow for estimation of an indicator relating to employee engagement in a global version of the index.

- Metrics table: *not available*
- Framework definition: Section 2.4.1

8.4.2 Diversity and inclusion metrics

There are four sources for the diversity and inclusion theme: the EIGE Gender Statistics Database (EIGE), the International Labor Organization (ILO), the International Survey of Revenue Administration (ISORA), and the Varieties of Democracy Dataset (VDEM). Ten measures are extracted from these sources to estimate three indicators: inclusive recruitment; gender diversity (all staff); and, gender diversity (management).

The inclusive recruitment indicator measures the extent to which individuals with different characteristics or backgrounds are able to apply for government jobs. It is estimated from the VDEM measures on whether state jobs are open to those regardless of their gender, their rural or urban location, their socio-economic position, their social group (e.g. ethnic or religious group), or their political group.

The gender diversity (all staff) indicator measures the extent to which gender parity is achieved across the employee base. It is estimated from data extracted from the ILO on women in public administration employment, the ISORA measure on the proportion of all those working at the tax agency that are women, and data extracted from the ILO comparing the hourly earnings of men and women in the public sector.

The gender diversity (management) indicator measures the extent to which gender parity is achieved at management/senior levels of the public sector. It is estimated from data extracted from the EIGE and ILO on women in management positions, and the ISORA measure on the proportion of senior officials working at the tax agency that are women.

An additional metric was identified from OECD/EU only sources that would provide an additional metrics for the estimate of inclusive recruitment.

- Metrics table: Table 8.14
- Framework definition: Section 2.4.2

8.4.3 HR management

There are two sources for the HR management theme: the Quality of Government Expert Survey (QOG) and the Varieties of Democracy Dataset (VDEM). Five measures are extracted from these sources to estimate two indicators: recruitment regime; and, meritocratic appointment.

The recruitment regime indicator measures the openness and security of recruitment to public administration jobs. It is estimated by the QOG measures on hierarchy levels subject to open recruitment and on the security of tenure enjoyed by public employees.

The meritocratic appointment indicator measures the extent to which appointment decisions in the public administration are based on merit and not on patronage. It is estimated by the QOG measures on recruitment by merit and patronage, and the VDEM measure on recruitment not by patronage.

- Metrics table: Table 8.15
- Framework definition: Section 2.4.3

8.4.4 Procurement

There is one source for the procurement theme: the Global Data Barometer (GDB). One measure is extracted from this source to estimate a single indicator: procurement data.

The procurement data indicator measures the availability of data relating to public procurement tenders and contracts. It is estimated from the GDB measure on procurement data.

- Metrics table: Table 8.16
- Framework definition: Section 2.4.4

8.4.5 Technology and workplaces metrics

There is one source for the technology and workplaces theme: the GovTech Maturity Index (GTMI). One measure is extracted from this source to estimate a single indicator: administrative IT systems.

The administrative IT systems indicator measures the existence of management information systems for administrative activities. It is estimated based on data extracted from the GTMI on the existence and nature of MI systems for public finances, HR management, payroll services and public procurement.

- Metrics table: Table 8.17
- Framework definition: Section 2.4.5

8.5 Metrics definition tables

The following tables provide metadata about the metrics for each of the 4 domains and 16 themes that comprise the Blavatnik Index of Public Administration 2024. For each metric the table provides a label, the data source reference, the original variable label in the data source, the type of collection used by the data source, the reference year(s) of the data used by the Blavatnik Index, any transformation applied to the data, and the number of countries for which there is data available.

Table 8.2: Metrics for the strategic capacity theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Prioritisation	Setting priorities	BTI	To what extent does the government set and maintain strategic priorities?	Opinion - expert	2023	BTI & SGI rescaling	137
	Resource efficiency	BTI	To what extent does the government make efficient use of available human, financial and organizational resources?	Opinion - expert	2023	BTI & SGI rescaling	137
	Influence of strategy units	SGI	How much influence do strategic planning units and bodies have on government decision-making?	Opinion - expert	2022	BTI & SGI rescaling	41
	Reviewing policy priorities	SGI	Does the government office/prime minister’s office have the expertise to evaluate ministerial draft bills according to the government’s priorities?	Opinion - expert	2022	BTI & SGI rescaling	41

Table 8.3: Metrics for the openness and communications theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Right to information	Right to information	ROLI	Right to information	Opinion - expert & general public	2023	None	142
Open government	Open government and data portals	GTMI	Open government and data portals	Government self-assessment	2022	None	198
	Published laws	ROLI	Publicized laws and government data	Opinion - expert & general public	2023	None	142
Engagement and feedback	Consultation with social actors	SGI	Does the government consult with societal actors in a fair and pluralistic manner?	Opinion - expert	2022	BTI & SGI rescaling	41
	Consultation with civil society	BTI	To what extent does the political leadership consult civil society actors in their plurality in policymaking?	Opinion - expert	2023	BTI & SGI rescaling	137
	Complaint mechanisms	ROLI	Complaint mechanisms	Opinion - expert & general public	2023	None	142
	Digital participation methods	GTMI	Digital methods for public participation	Government self-assessment	2022	None	198

Table 8.4: Metrics for the integrity theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Impartial behaviour	Policy implementation is impartial	QOG	In your chosen country, how often do bureaucrats of the central government act impartially when implementing laws and regulations?	Opinion - expert	2020	None	84
	Decisions free of interference	QOG	In your chosen country, how often is day-to-day bureaucratic decision-making subject to interference from individual politicians?	Opinion - expert	2020	Inverted	83
	Respect for due process	ROLI	Due process is respected in administrative proceedings	Opinion - expert & general public	2023	None	142
	Officials impartial in their duties	VDEM	Are public officials rigorous and impartial in the performance of their duties?	Opinion - expert	2023	None	179
Corruption	Officials involved in corruption	GCB	How many of the following people do you think are involved in corruption or haven't you heard enough about them to say: Government officials	Opinion - general public	2019-2021	Inverted	110
	Officials accept bribes	VDEM	How routinely do public sector employees grant favors in exchange for bribes, kickbacks, or other material inducements?	Opinion - expert	2023	None	179
	Officials misuse resources	VDEM	How often do public sector employees steal, embezzle, or misappropriate public funds or other state resources for personal or family use?	Opinion - expert	2023	None	179
Sanctions	Officials prosecuted for abuse	BTI	To what extent are public officeholders who abuse their positions prosecuted or penalized?	Opinion - expert	2023	None	137
	Officials sanctioned for misconduct	ROLI	Government officials are sanctioned for misconduct	Opinion - expert & general public	2023	None	142
Integrity data	Data for public integrity	GDB	Integrity data availability and impact	Expert assessment	2021	None	109

Table 8.5: Metrics for the innovation theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Innovative approach	Innovative and flexible approach	BTI	How innovative and flexible is the government?	Opinion - expert	2023	None	137
	Digital innovation support	GTMI	Support for digital innovations within government	Government self-assessment	2022	None	198
	Innovative tax agency	ISORA	Use of innovative practices and technologies by the tax administration	Government self-assessment	2019-2021	None	172

Table 8.6: Metrics for the policmaking theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Policy coordination	Coordinating policy proposals	SGI	How effectively do ministerial or cabinet committees coordinate cabinet proposals?	Opinion - expert	2022	BTI & SGI rescaling	41
	Coordinating conflicting objectives	BTI	To what extent can the government coordinate conflicting objectives into a coherent policy?	Opinion - expert	2023	BTI & SGI rescaling	137

Table 8.7: Metrics for the regulation theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Regulatory enforcement	Regulations enforced effectively	ROLI	Government regulations are effectively enforced	Opinion - expert & general public	2023	None	142
	Regulations enforced properly	ROLI	Government regulations are applied and enforced without improper influence	Opinion - expert & general public	2023	None	142

Table 8.8: Metrics for the crisis and risk management

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Natural hazards	Disaster risk reduction strategies	SFM	Disaster risk reduction strategies	Government self-assessment	2019-2023	None	115
Cybersecurity	Cybersecurity capacity	GCI	Capacity building measures	Government self-assessment	2020	None	190
	Cybersecurity cooperation	GCI	Cooperative measures	Government self-assessment	2020	None	188
	Cybersecurity legal measures	GCI	Legal measures	Government self-assessment	2020	None	191
	Cybersecurity organisational measures	GCI	Organisational measures	Government self-assessment	2020	None	191
	Cybersecurity technical measures	GCI	Technical measures	Government self-assessment	2020	None	191

Table 8.9: Metrics for the use of data and statistics theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Data capability	Governance of data and statistics	GDB	Data governance	Expert assessment	2021	None	109
	Data capability	GDB	Data capability	Expert assessment	2021	None	109
Data availability	Economic statistics availability	ODIN	Economic statistics openness	Expert assessment	2022	None	195
	Company data availability	GDB	Company data availability and impact	Expert assessment	2021	None	109
	Land data availability	GDB	Land data availability and impact	Expert assessment	2021	None	109
	Social statistics availability	ODIN	Social statistics openness	Expert assessment	2022	None	195
	Health data availability	GDB	Health data availability	Expert assessment	2021	None	109
	Environmental statistics availability	ODIN	Environmental statistics openness	Expert assessment	2022	None	195
	Climate data availability	GDB	Climate data availability	Expert assessment	2021	None	109
Data coverage	Economic statistics coverage	ODIN	Economic statistics coverage	Expert assessment	2022	None	195
	Environmental statistics coverage	ODIN	Environmental statistics coverage	Expert assessment	2022	None	195
	Social statistics coverage	ODIN	Social statistics coverage	Expert assessment	2022	None	195
Data re-use	Use of statistics in policy documents	PARIS21	Use of statistics in national policy documents	Statistical analysis	2019-2020	None	183

Table 8.10: Metrics for the system oversight theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Policy implementation	Achieve policy objectives	SGI	To what extent can the government achieve its own policy objectives?	Opinion - expert	2022	BTI & SGI rescaling	41
	Implement policy objectives	BTI	How effective is the government in implementing its own policies?	Opinion - expert	2023	BTI & SGI rescaling	137

Table 8.11: Metrics for the digital services theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Digital strategy & policy	Digital strategy & policy	GTMI	Strategies and policies for digital government	Government self-assessment	2022	None	198
Infrastructure	Technology and infrastructure	GTMI	Back-end technologies and infrastructure	Government self-assessment	2022	None	198
Digital public services	End-user services	GTMI	Digital services and technologies for end-users (portals, e-payment, digital signatures)	Government self-assessment	2022	None	198

Table 8.12: Metrics for the tax administration theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Tax agency administration	Administrative costs	ISORA	Administrative costs (% of revenue)	Administrative data	2019-2021	Inverted	159
	Turnover of tax agency staff	ISORA	Turnover of tax agency staff	Administrative data	2019-2021	Median distance	165
Tax compliance	Corporate taxes filed on-time	ISORA	Corporate tax (% filed on-time)	Administrative data	2019-2021	None	152
	Personal taxes filed on-time	ISORA	Personal tax (% filed on-time)	Administrative data	2019-2021	None	143
	Corporate taxes paid on-time	ISORA	Corporate tax (% paid on-time)	Administrative data	2019-2021	None	119
	Personal taxes paid on-time	ISORA	Personal tax (% paid on-time)	Administrative data	2019-2021	None	111
	Time to file taxes	DB	Time to file business taxes	Administrative data	2020	Inverted	180
	Tax debts/arrears	ISORA	Tax debt (% of revenue)	Administrative data	2019-2021	Inverted	152
Digital tax services	Corporate taxes filed electronically	ISORA	Electronic filing: corporate tax	Administrative data	2019-2021	None	142
	Personal taxes paid electronically	ISORA	Electronic filing: personal tax	Administrative data	2019-2021	None	135
	Taxes paid electronically	ISORA	Electronic payments (%)	Administrative data	2019-2021	None	156
	Digital channels for tax advice/help	ISORA	Proportion of service contacts via digital channels	Administrative data	2019-2021	None	129

Table 8.13: Metrics for the border services theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Customs processes	Efficiency of customs processes	LPI	Efficiency of customs processes	Opinion - businesses	2022	None	138

Table 8.14: Metrics for the diversity and inclusion theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Inclusive recruitment	Inclusive recruitment: gender	VDEM	Are state jobs equally open to qualified individuals regardless of gender?	Opinion - expert	2023	None	176
	Inclusive recruitment: location	VDEM	Are state jobs equally open to qualified individuals regardless of their rural or urban location?	Opinion - expert	2023	None	176
	Inclusive recruitment: socio-economic background	VDEM	Are state jobs equally open to qualified individuals regardless of socio-economic position?	Opinion - expert	2023	None	176
	Inclusive recruitment: social group	VDEM	Are state jobs equally open to qualified individuals regardless of social group?	Opinion - expert	2023	None	176
	Inclusive recruitment: political affiliation	VDEM	Are state jobs equally open to qualified individuals regardless of their association with a political group?	Opinion - expert	2023	None	176
Gender diversity (all staff)	Women in public administration (rate)	ILO	Public administration employment: percent female	Official statistics	2019-2023	Distance from 50%	154
	Women in tax agency workforce	ISORA	Proportion of all tax officials that are female	Government self-assessment	2019-2021	Distance from 50%	170
	Ratio of female to male public-sector pay	ILO	Public sector pay ratio: female-to-male	Official statistics	2019-2023	Distance from 1	68
Gender diversity (management)	Women in public sector management	GSD/ILO	Public sector managers: percent female	Official statistics	2019-2023	Distance from 50%	111
	Women in senior tax agency roles	ISORA	Proportion of senior tax officials that are female	Government self-assessment	2019-2021	Distance from 50%	163

Table 8.15: Metrics for the HR management theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Recruitment regime	Recruitment level of entry	QOG	On average, do most ministries and other bureaucratic agencies of the central government in your chosen country resemble closed or open type of organizations?	Opinion - expert	2020	Inverted	110
	Security of tenure	QOG	In your chosen country, to what extent do bureaucrats working in ministries and other bureaucratic agencies of the central government enjoy security of tenure?	Opinion - expert	2020	None	90
Meritocratic appointment	Recruitment by merit	QOG	In your chosen country, to what extent are appointments to bureaucratic positions in the central government based on individuals' merits - such as knowledge, skills and job-related experience?	Opinion - expert	2020	None	86
	Recruitment by patronage (QOG)	QOG	In your chosen country, to what extent are appointments to bureaucratic positions in the central government based on the political and/or personal connections of the applicant?	Opinion - expert	2020	Inverted	90
	Recruitment not by patronage (VDEM)	VDEM	To what extent are appointment decisions in the state administration [not] based on personal and political connections, as opposed to skills and merit?	Opinion - expert	2023	None	179

Table 8.16: Metrics for the procurement theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Procurement data	Public procurement data	GDB	Procurement data and impact	Expert assessment	2021	None	109

Table 8.17: Metrics for the technology and workplaces theme

Indicator	Metric	Source	Source variable	Type of collection	Year(s)	Data transformation	Number of countries
Administrative IT systems	Administrative IT systems	GTMI	Administrative IT systems (finance, HR, payroll and procurement systems)	Government self-assessment	2022	None	198

9 Data coverage and country selection

Any analytical exercise needs to carefully consider the quality of its data, this is particularly important when combining data from different data sources. This section explores country data quality, in particular data availability for countries, to determine inclusion in the Index's calculations.

Of the 212 countries and territories which have data for at least one of the 86 metrics used for the global index, only one country (Mexico) has data for all 86 metrics, as a result an approach is needed to consider a country's data coverage for determining its inclusion in the calculation of the Index.

A simple approach for determining coverage would be to consider simply the total number of metrics a country has data for (m_c) as a proportion of the total number of possible metrics (M).

$$p_m = \frac{m_c}{M}$$

However, this approach does not consider how a country's data is spread across the structure of our conceptual framework and data model – a country with a higher proportion of metrics concentrated in fewer themes would do better on this simple measure than a country with a lower proportion of metrics spread across more themes. As described in section 6 missing data is not subject to imputation, good coverage of the data model is therefore an important consideration to ensure that the implicit estimate of missing data will come from "sibling" components in the data model.

Based on the chi-square test we have developed an algorithm for assessing a country's data availability. This algorithm assesses both the volume of metrics a country has as well as the spread of that data over the data model. For each country-theme combination, an information quotient is calculated as the square of the difference between the total number of metrics a country has for that theme ($m_{t,c}$) and the number of indicators a country has for that theme ($i_{t,c}$) divided by the overall number of indicators in that theme (i_t). The overall data coverage score for a country (Q_c) is calculated as the sum of these country-theme information quotients adjusted for overall coverage across the themes, calculated as the number of themes a country has data for (t_c) divided by overall number of themes (T).

$$Q_c = \sum_{t=1}^{t_c} \frac{(m_{t,c} - i_{t,c} + 1)^2}{i_t}$$

Having calculated the data coverage algorithm, we can set criteria to determine inclusion in the calculation of the Index and its components. Two principal criteria were selected :

- The overall data coverage score (Q_c) is greater than or equal to half its theoretical maximum value (i.e. the score a country would achieve if it has data for all metrics). The maximum data coverage score is 161.08, the threshold value is therefore 80.54.
- While the overall score does take into consideration the number of metrics, the overall percentage of metrics a country has (p_m) is also retained as a secondary threshold. It was decided to only include countries and territories with at least 2/3rds of overall metrics.

These criteria give rise to a selection of 120 countries and territories[1].

The inclusion criteria provide a binary state of whether countries are included or not however we can also devise a grading scheme using the same measures to determine inclusion to give a better sense of an individual country's data coverage. For countries and territories above thresholds for inclusion in the Index the lower quartile, median and upper quartile of their data coverage score and percentage of metrics are used to define the boundaries for the first four grades (A-D); for countries and territories below the thresholds for inclusion in the Index the median of their data coverage score and percentage of metrics is used to define the boundary between the fifth and sixth grade (E-F). The following tables provide summarises of data coverage metrics: Table 9.1 by grade; Table 9.2 by regional, economic and population groupings. Full data coverage metrics for all countries and territories are provided in Table 9.3 and Table 9.4.

[1] Hong Kong appears separately in many of the data sources used to calculate the Blavatnik Index and while it was above the thresholds for inclusion with a data coverage score of 98.92 and 74.4% of total metrics, it has not been included as it is a territory of China rather than a separate national entity. As a result in Table 9.1, Hong Kong has been included in grade E, however Hong Kong's underlying data coverage metrics are equivalent to other countries classified as grade D.

Table 9.1: Summary of data coverage metrics by grade

Data coverage grade	Data coverage thresholds[1]	Number of countries
Countries included in the index		
A	Qc >= 127.50, pm >= 90.24%	30
B	Qc >= 116.17, pm >= 85.37%	27
C	Qc >= 102.75, pm >= 80.49%	28
D	Qc >= 80.54, pm >= 66.67%	35
Countries and territories not included in the index		
E[2]	Qc >= 53.25, pm >= 48.78%	46
F	Qc > 0, pm > 0%	46
Ungraded (no data)	Qc = 0, pm = 0%	43

[1]Qc is the data coverage score, pm is the percent of total metrics
[2]Hong Kong included in many of the source datasets used by the Index and has Qc and pm scores equivalent to grade D. As a territory of China, Hong Kong is not included in the Blavatnik Index, therefore it has been included in this summary table in the total number of countries for grade E.

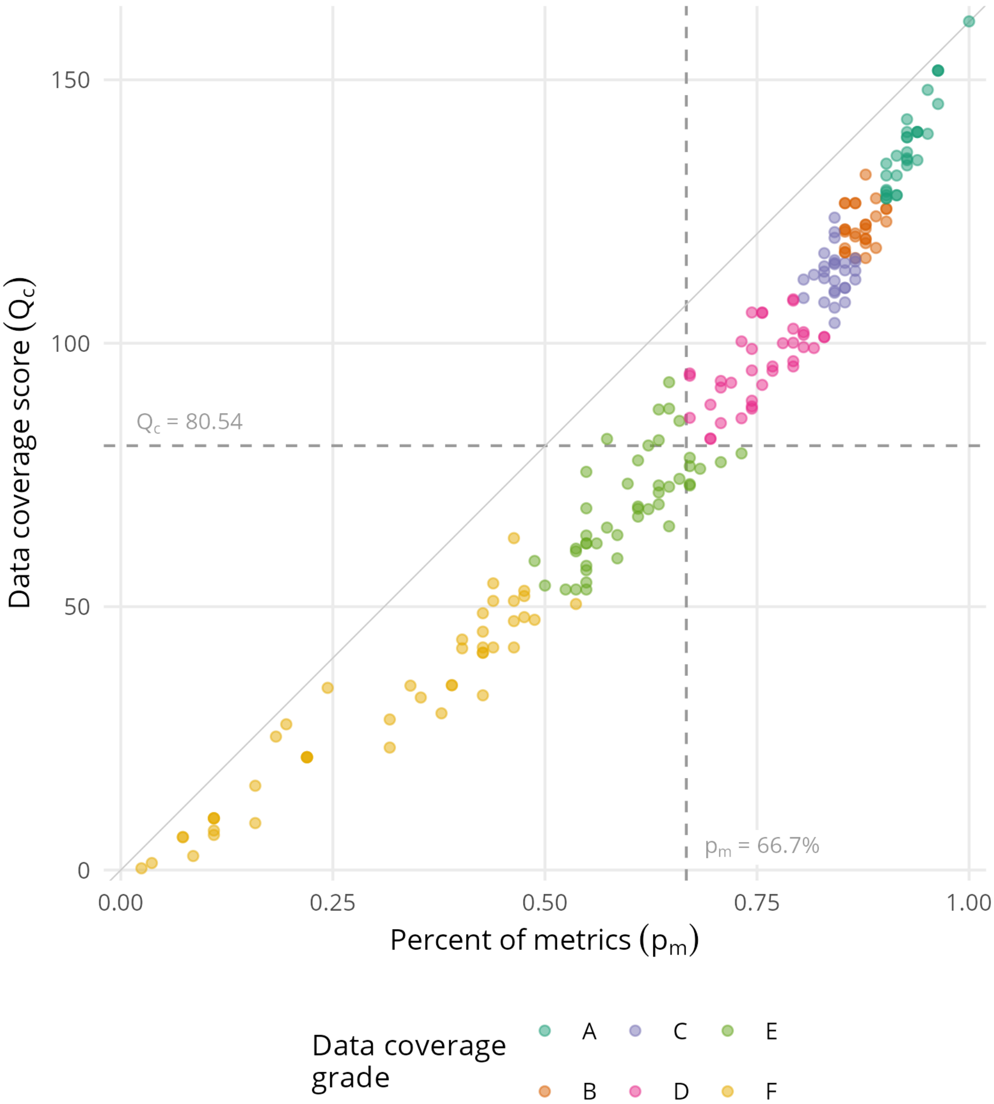
Table 9.2: Summary of country coverage by groupings

	Countries included in the Index	UN members not included	Other entities not included[1]
All countries and territories	120	74	60
Geographic region			
Americas	22	13	22
Asia and Pacific	18	19	18
Eastern Europe	20	2	1
Middle East, North Africa & Central Asia	15	14	4
Sub-Saharan Africa	29	19	7
Western Europe	16	7	8
Income group (World Bank, 2023)			
High income	39	21	24
Low income	13	13	—
Lower middle income	33	21	—
Upper middle income	35	18	1
No World Bank income classification	—	1	35
Population size (World Bank, 2022)			
Less than 5 million	26	46	20
5 million to 10 million	23	7	2
10 million to 25 million	27	9	—
25 million to 50 million	20	7	—
50 million or greater	24	5	—
No population data	—	—	38

[1]Other entities includes countries that are not UN members, dependent and other territories of other countries, disputed countries and territories, or those with limited recognition.

As shown in Figure 9.1, while closely correlated there is a slightly non-linear relationship between the data coverage score and the percent of metrics measures, this is expected since the data coverage score applies a penalty for a country not having complete data across the measurement framework.

Figure 9.1: Scatter plot of data coverage measures for all countries and territories



There is no statistically significant correlation between a country's overall score in the Blavatnik Index and either its data coverage score or the percent of metrics it has data for. However, as Figure 9.2 shows, there does appear a relationship between the minimum score for the Blavatnik Index and the data coverage score[2], which is likely a function of the data normalisation process that results in country scores becoming relative measures between 0 and 1.

[2]This effect is also apparent when looking at the percent of total metrics.

Figure 9.2: Scatter plot of the data coverage score and Blavatnik Index score

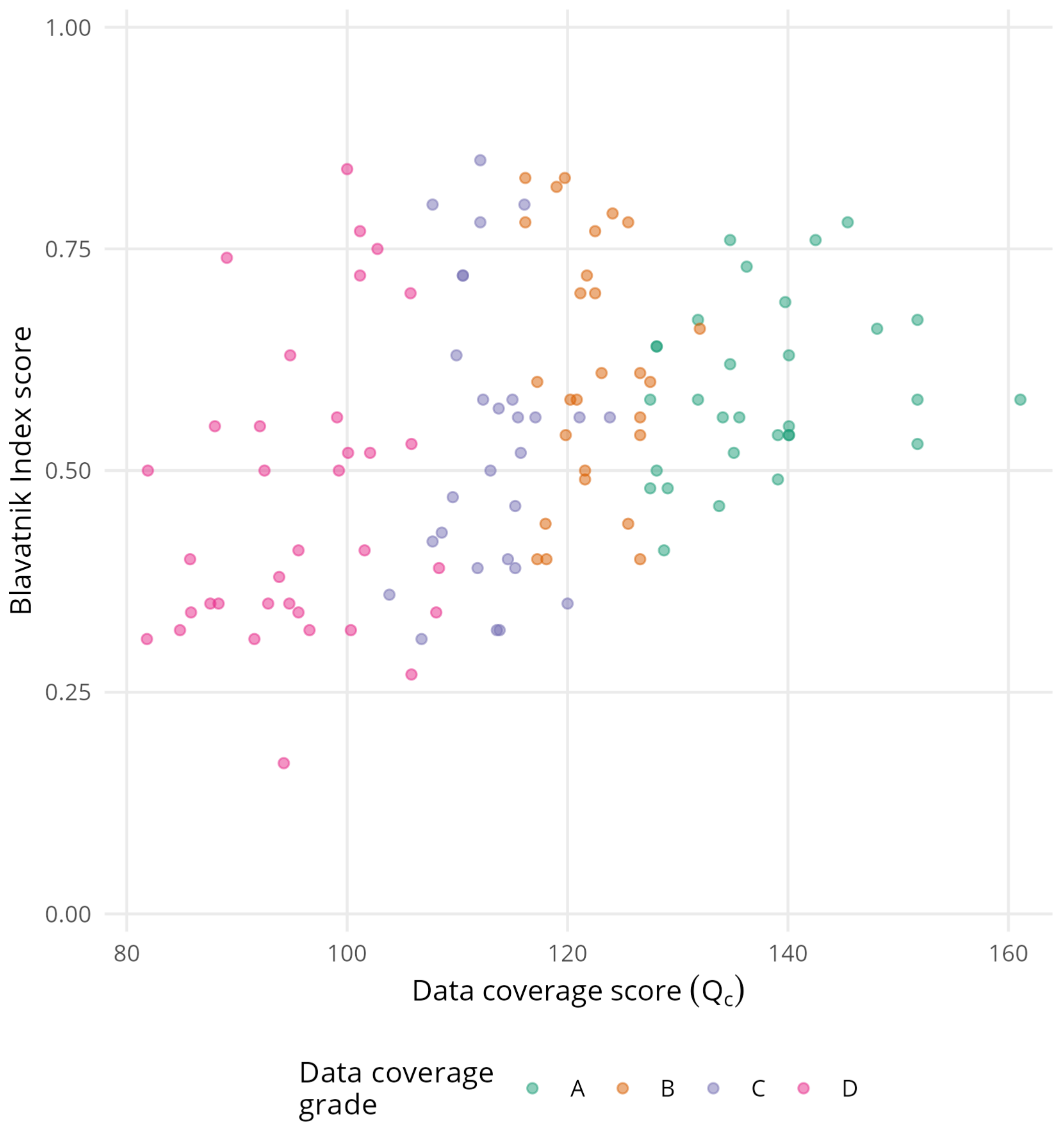


Table 9.3: Data coverage metadata for countries included in the Blavatnik Index of public administration 2024

					Data coverage		
Country	Code	Status	Region	Income group	Grade	Score	% metrics
Albania	ALB	UN member	Eastern Europe	Upper middle income	A	131.83	91.5%
Angola	AGO	UN member	Sub-Saharan Africa	Lower middle income	C	113.83	85.4%
Argentina	ARG	UN member	Americas	Upper middle income	A	140.08	92.7%
Armenia	ARM	UN member	Middle East, North Africa & Central Asia	Upper middle income	C	123.83	84.1%
Australia	AUS	UN member	Asia and Pacific	High income	B	124.08	89.0%
Austria	AUT	UN member	Western Europe	High income	D	89.08	74.4%
Azerbaijan	AZE	UN member	Middle East, North Africa & Central Asia	Upper middle income	D	92.50	72.0%
Bangladesh	BGD	UN member	Asia and Pacific	Lower middle income	B	118.08	89.0%
Belarus	BLR	UN member	Eastern Europe	Upper middle income	C	111.83	84.1%
Belgium	BEL	UN member	Western Europe	High income	D	102.75	79.3%
Benin	BEN	UN member	Sub-Saharan Africa	Lower middle income	B	121.58	85.4%
Bolivia	BOL	UN member	Americas	Lower middle income	B	118.00	85.4%
Botswana	BWA	UN member	Sub-Saharan Africa	Upper middle income	A	128.08	91.5%
Brazil	BRA	UN member	Americas	Upper middle income	A	128.08	90.2%
Bulgaria	BGR	UN member	Eastern Europe	Upper middle income	A	151.75	96.3%
Burkina Faso	BFA	UN member	Sub-Saharan Africa	Low income	C	115.25	84.1%
Cambodia	KHM	UN member	Asia and Pacific	Lower middle income	C	106.75	84.1%
Cameroon	CMR	UN member	Sub-Saharan Africa	Lower middle income	C	103.83	84.1%
Canada	CAN	UN member	Americas	High income	B	119.75	87.8%
Chile	CHL	UN member	Americas	High income	A	139.75	95.1%
China	CHN	UN member	Asia and Pacific	Upper middle income	D	99.08	81.7%
Colombia	COL	UN member	Americas	Upper middle income	A	128.08	91.5%
Costa Rica	CRI	UN member	Americas	Upper middle income	A	140.08	93.9%
Croatia	HRV	UN member	Eastern Europe	High income	A	134.75	92.7%
Czechia	CZE	UN member	Eastern Europe	High income	A	151.75	96.3%
Denmark	DNK	UN member	Western Europe	High income	B	116.17	87.8%
Dominican Republic	DOM	UN member	Americas	Upper middle income	B	126.58	85.4%
Ecuador	ECU	UN member	Americas	Upper middle income	C	115.75	84.1%
El Salvador	SLV	UN member	Americas	Upper middle income	C	114.58	82.9%
Estonia	EST	UN member	Eastern Europe	High income	A	145.42	96.3%
Eswatini	SWZ	UN member	Sub-Saharan Africa	Lower middle income	D	93.83	67.1%
Ethiopia	ETH	UN member	Sub-Saharan Africa	Low income	D	85.75	73.2%
Finland	FIN	UN member	Western Europe	High income	B	119.00	87.8%
France	FRA	UN member	Western Europe	High income	B	125.50	90.2%
Gabon	GAB	UN member	Sub-Saharan Africa	Upper middle income	D	81.83	69.5%
Gambia	GMB	UN member	Sub-Saharan Africa	Low income	D	96.58	79.3%
Georgia	GEO	UN member	Middle East, North Africa & Central Asia	Upper middle income	B	120.83	86.6%
Germany	DEU	UN member	Western Europe	High income	D	101.17	82.9%

Ghana	GHA	UN member	Sub-Saharan Africa	Lower middle income	A	133.75	92.7%
Greece	GRC	UN member	Western Europe	High income	A	131.83	90.2%
Guatemala	GTM	UN member	Americas	Upper middle income	C	115.25	85.4%
Guinea	GIN	UN member	Sub-Saharan Africa	Lower middle income	D	85.83	67.1%
Guyana	GUY	UN member	Americas	High income	C	120.00	84.1%
Honduras	HND	UN member	Americas	Lower middle income	B	126.58	85.4%
Hungary	HUN	UN member	Eastern Europe	High income	C	115.00	84.1%
India	IND	UN member	Asia and Pacific	Lower middle income	C	113.75	86.6%
Indonesia	IDN	UN member	Asia and Pacific	Upper middle income	B	123.08	90.2%
Ireland	IRL	UN member	Western Europe	High income	B	122.50	87.8%
Israel	ISR	UN member	Middle East, North Africa & Central Asia	High income	D	105.75	75.6%
Italy	ITA	UN member	Western Europe	High income	D	101.17	82.9%
Jamaica	JAM	UN member	Americas	Upper middle income	A	129.08	90.2%
Jordan	JOR	UN member	Middle East, North Africa & Central Asia	Lower middle income	A	135.58	91.5%
Kazakhstan	KAZ	UN member	Middle East, North Africa & Central Asia	Upper middle income	B	127.50	89.0%
Kenya	KEN	UN member	Sub-Saharan Africa	Lower middle income	A	139.08	92.7%
Kosovo	XKK	Not an ISO 3166-1 entitty	Eastern Europe	Upper middle income	D	105.83	74.4%
Kyrgyzstan	KGZ	UN member	Middle East, North Africa & Central Asia	Lower middle income	B	125.50	90.2%
Latvia	LVA	UN member	Eastern Europe	High income	A	136.25	92.7%
Lebanon	LBN	UN member	Middle East, North Africa & Central Asia	Lower middle income	D	94.75	76.8%
Lesotho	LSO	UN member	Sub-Saharan Africa	Lower middle income	D	88.33	69.5%
Liberia	LBR	UN member	Sub-Saharan Africa	Low income	C	113.58	82.9%
Lithuania	LTU	UN member	Eastern Europe	High income	A	142.50	92.7%
Madagascar	MDG	UN member	Sub-Saharan Africa	Low income	D	84.83	70.7%
Malawi	MWI	UN member	Sub-Saharan Africa	Low income	D	95.58	79.3%
Malaysia	MYS	UN member	Asia and Pacific	Upper middle income	B	117.25	85.4%
Malta	MLT	UN member	Western Europe	High income	C	109.92	84.1%
Mauritius	MUS	UN member	Sub-Saharan Africa	Upper middle income	D	94.83	74.4%
Mexico	MEX	UN member	Americas	Upper middle income	A	161.08	100.0%
Moldova	MDA	UN member	Eastern Europe	Upper middle income	C	112.33	82.9%
Mongolia	MNG	UN member	Asia and Pacific	Lower middle income	B	126.58	86.6%
Montenegro	MNE	UN member	Eastern Europe	Upper middle income	D	100.08	79.3%
Morocco	MAR	UN member	Middle East, North Africa & Central Asia	Lower middle income	D	99.25	80.5%
Mozambique	MOZ	UN member	Sub-Saharan Africa	Low income	D	108.08	79.3%
Myanmar	MMR	UN member	Asia and Pacific	Lower middle income	D	105.83	75.6%
Namibia	NAM	UN member	Sub-Saharan Africa	Upper middle income	D	101.58	80.5%
Nepal	NPL	UN member	Asia and Pacific	Lower middle income	C	107.75	82.9%
Netherlands	NLD	UN member	Western Europe	High income	B	122.50	87.8%
New Zealand	NZL	UN member	Asia and Pacific	High income	C	116.08	86.6%
Nicaragua	NIC	UN member	Americas	Lower middle income	D	91.58	70.7%
Niger	NER	UN member	Sub-Saharan Africa	Low income	D	92.83	70.7%
Nigeria	NGA	UN member	Sub-Saharan Africa	Lower middle income	A	128.75	90.2%
North Macedonia	MKD	UN member	Eastern Europe	Upper middle income	D	88.00	74.4%
Norway	NOR	UN member	Western Europe	High income	D	100.00	78.0%

Pakistan	PAK	UN member	Asia and Pacific	Lower middle income	D	95.58	76.8%
Panama	PAN	UN member	Americas	High income	B	126.58	86.6%
Paraguay	PRY	UN member	Americas	Upper middle income	A	134.08	90.2%
Peru	PER	UN member	Americas	Upper middle income	A	140.08	93.9%
Philippines	PHL	UN member	Asia and Pacific	Lower middle income	A	140.08	93.9%
Poland	POL	UN member	Eastern Europe	High income	B	132.00	87.8%
Portugal	PRT	UN member	Western Europe	High income	C	110.50	85.4%
Romania	ROU	UN member	Eastern Europe	High income	A	151.75	96.3%
Russia	RUS	UN member	Eastern Europe	Upper middle income	C	115.50	86.6%
Rwanda	RWA	UN member	Sub-Saharan Africa	Low income	D	102.08	80.5%
Saudi Arabia	SAU	UN member	Middle East, North Africa & Central Asia	High income	D	92.08	75.6%
Senegal	SEN	UN member	Sub-Saharan Africa	Lower middle income	C	108.58	80.5%
Serbia	SRB	UN member	Eastern Europe	Upper middle income	C	117.08	82.9%
Sierra Leone	SLE	UN member	Sub-Saharan Africa	Low income	D	87.58	74.4%
Singapore	SGP	UN member	Asia and Pacific	High income	C	112.08	80.5%
Slovakia	SVK	UN member	Eastern Europe	High income	A	148.08	95.1%
Slovenia	SVN	UN member	Eastern Europe	High income	B	121.17	85.4%
South Africa	ZAF	UN member	Sub-Saharan Africa	Upper middle income	A	135.08	92.7%
South Korea	KOR	UN member	Asia and Pacific	High income	A	134.75	93.9%
Spain	ESP	UN member	Western Europe	High income	B	116.17	86.6%
Sri Lanka	LKA	UN member	Asia and Pacific	Lower middle income	C	109.58	84.1%
Sudan	SDN	UN member	Middle East, North Africa & Central Asia	Low income	D	94.25	67.1%
Sweden	SWE	UN member	Western Europe	High income	C	110.50	85.4%
Tajikistan	TJK	UN member	Middle East, North Africa & Central Asia	Lower middle income	D	100.33	73.2%
Thailand	THA	UN member	Asia and Pacific	Upper middle income	B	120.25	86.6%
Togo	TGO	UN member	Sub-Saharan Africa	Low income	B	117.25	85.4%
Trinidad & Tobago	TTO	UN member	Americas	High income	B	121.58	85.4%
Tunisia	TUN	UN member	Middle East, North Africa & Central Asia	Lower middle income	D	81.92	69.5%
Türkiye	TUR	UN member	Middle East, North Africa & Central Asia	Upper middle income	C	121.08	84.1%
Uganda	UGA	UN member	Sub-Saharan Africa	Low income	A	139.08	92.7%
Ukraine	UKR	UN member	Eastern Europe	Lower middle income	A	127.50	90.2%
United Kingdom	GBR	UN member	Western Europe	High income	C	107.75	85.4%
United States	USA	UN member	Americas	High income	C	112.08	86.6%
Uruguay	URY	UN member	Americas	High income	B	121.75	87.8%
Uzbekistan	UZB	UN member	Middle East, North Africa & Central Asia	Lower middle income	A	127.50	90.2%
Vietnam	VNM	UN member	Asia and Pacific	Lower middle income	B	119.83	87.8%
Zambia	ZMB	UN member	Sub-Saharan Africa	Lower middle income	C	113.00	81.7%
Zimbabwe	ZWE	UN member	Sub-Saharan Africa	Lower middle income	D	108.33	79.3%

Table 9.4: Data coverage metadata for countries and territories not included in the Blavatnik Index of public administration 2024

					Data coverage		
Country/Territory	Code	Status	Region	Income group	Grade	Score	% metrics
Afghanistan	AFG	UN member	Middle East, North Africa & Central Asia	Low income	E	77.42	70.7%
Åland	ALA	Other ISO 3166-1 entity	Western Europe	—	—	—	—
Algeria	DZA	UN member	Middle East, North Africa & Central Asia	Lower middle income	E	53.25	53.7%
American Samoa	ASM	Other ISO 3166-1 entity	Asia and Pacific	High income	—	—	—
Andorra	AND	UN member	Western Europe	High income	F	21.42	22.0%
Anguilla	AIA	Other ISO 3166-1 entity	Americas	—	F	6.25	7.3%
Antigua & Barbuda	ATG	UN member	Americas	High income	E	54.00	50.0%
Aruba	ABW	Other ISO 3166-1 entity	Americas	High income	F	34.58	24.4%
Bahamas	BHS	UN member	Americas	High income	F	51.08	46.3%
Bahrain	BHR	UN member	Middle East, North Africa & Central Asia	High income	E	62.00	54.9%
Barbados	BRB	UN member	Americas	High income	E	85.25	65.9%
Belize	BLZ	UN member	Americas	Upper middle income	E	77.75	61.0%
Bermuda	BMU	Other ISO 3166-1 entity	Americas	High income	F	2.67	8.5%
Bhutan	BTN	UN member	Asia and Pacific	Lower middle income	E	81.58	63.4%
Bosnia & Herzegovina	BIH	UN member	Eastern Europe	Upper middle income	E	69.42	63.4%
Bouvet Island	BVT	Other ISO 3166-1 entity	Americas	—	—	—	—
British Indian Ocean Territory	IOT	Other ISO 3166-1 entity	Sub-Saharan Africa	—	—	—	—
British Virgin Islands	VGB	Other ISO 3166-1 entity	Americas	High income	—	—	—
Brunei	BRN	UN member	Asia and Pacific	High income	F	32.75	35.4%
Burundi	BDI	UN member	Sub-Saharan Africa	Low income	E	87.58	64.6%
Cabo Verde	CPV	UN member	Sub-Saharan Africa	Lower middle income	E	60.50	53.7%
Caribbean Netherlands	BES	Other ISO 3166-1 entity	Americas	—	—	—	—
Cayman Islands	CYM	Other ISO 3166-1 entity	Americas	High income	—	—	—
Central African Republic	CAF	UN member	Sub-Saharan Africa	Low income	E	80.58	62.2%
Chad	TCD	UN member	Sub-Saharan Africa	Low income	E	54.58	54.9%
Christmas Island	CXR	Other ISO 3166-1 entity	Asia and Pacific	—	—	—	—
Cocos (Keeling) Islands	CCK	Other ISO 3166-1 entity	Asia and Pacific	—	—	—	—
Comoros	COM	UN member	Sub-Saharan Africa	Lower middle income	F	63.00	46.3%
Congo (DRC)	COD	UN member	Sub-Saharan Africa	Low income	E	76.17	68.3%
Congo (ROC)	COG	UN member	Sub-Saharan Africa	Lower middle income	E	73.33	59.8%
Cook Islands	COK	Other ISO 3166-1 entity	Asia and Pacific	—	F	27.67	19.5%
Côte d'Ivoire	CIV	UN member	Sub-Saharan Africa	Lower middle income	E	74.25	65.9%
Cuba	CUB	UN member	Americas	Upper middle income	E	63.58	58.5%
Curaçao	CUW	Other ISO 3166-1 entity	Americas	High income	—	—	—
Cyprus	CYP	UN member	Eastern Europe	High income	E	73.25	67.1%
Djibouti	DJI	UN member	Sub-Saharan Africa	Lower middle income	F	42.25	43.9%
Dominica	DMA	UN member	Americas	Upper middle income	F	53.00	47.6%
Egypt	EGY	UN member	Middle East, North Africa & Central Asia	Lower middle income	E	76.67	67.1%

Equatorial Guinea	GNQ	UN member	Sub-Saharan Africa	Upper middle income	F	41.25	42.7%
Eritrea	ERI	UN member	Sub-Saharan Africa	Low income	F	43.75	40.2%
Falkland Islands	FLK	Other ISO 3166-1 entity	Americas	—	—	—	—
Faroe Islands	FRO	Other ISO 3166-1 entity	Western Europe	High income	—	—	—
Fiji	FJI	UN member	Asia and Pacific	Upper middle income	E	65.00	57.3%
French Guiana	GUF	Other ISO 3166-1 entity	Americas	—	—	—	—
French Polynesia	PYF	Other ISO 3166-1 entity	Asia and Pacific	High income	—	—	—
French Southern Territories	ATF	Other ISO 3166-1 entity	Sub-Saharan Africa	—	—	—	—
Gaza	XPG	Not an ISO 3166-1 entitty	Middle East, North Africa & Central Asia	—	F	9.83	11.0%
Gibraltar	GIB	Other ISO 3166-1 entity	Western Europe	High income	—	—	—
Greenland	GRL	Other ISO 3166-1 entity	Americas	High income	F	6.25	7.3%
Grenada	GRD	UN member	Americas	Upper middle income	E	68.67	54.9%
Guadeloupe	GLP	Other ISO 3166-1 entity	Americas	—	—	—	—
Guam	GUM	Other ISO 3166-1 entity	Asia and Pacific	High income	—	—	—
Guernsey	GGY	Other ISO 3166-1 entity	Western Europe	High income	—	—	—
Guinea-Bissau	GNB	UN member	Sub-Saharan Africa	Low income	F	52.00	47.6%
Haiti	HTI	UN member	Americas	Lower middle income	E	73.00	63.4%
Heard & McDonald Islands	HMD	Other ISO 3166-1 entity	Asia and Pacific	—	—	—	—
Hong Kong	HKG	Other ISO 3166-1 entity	Asia and Pacific	High income	D	98.92	74.4%
Iceland	ISL	UN member	Western Europe	High income	E	72.75	64.6%
Iran	IRN	UN member	Middle East, North Africa & Central Asia	Lower middle income	E	53.25	54.9%
Iraq	IRQ	UN member	Middle East, North Africa & Central Asia	Upper middle income	E	57.75	54.9%
Isle of Man	IMN	Other ISO 3166-1 entity	Western Europe	High income	—	—	—
Japan	JPN	UN member	Asia and Pacific	High income	E	79.08	73.2%
Jersey	JEY	Other ISO 3166-1 entity	Western Europe	High income	—	—	—
Kiribati	KIR	UN member	Asia and Pacific	Lower middle income	F	54.42	43.9%
Kuwait	KWT	UN member	Middle East, North Africa & Central Asia	High income	E	53.25	52.4%
Laos	LAO	UN member	Asia and Pacific	Lower middle income	E	68.58	61.0%
Libya	LBY	UN member	Middle East, North Africa & Central Asia	Upper middle income	F	47.25	46.3%
Liechtenstein	LIE	UN member	Western Europe	High income	F	21.42	22.0%
Luxembourg	LUX	UN member	Western Europe	High income	E	65.25	64.6%
Macau	MAC	Other ISO 3166-1 entity	Asia and Pacific	High income	F	8.92	15.9%
Maldives	MDV	UN member	Asia and Pacific	Upper middle income	E	81.83	57.3%
Mali	MLI	UN member	Sub-Saharan Africa	Low income	E	59.17	58.5%
Marshall Islands	MHL	UN member	Asia and Pacific	Upper middle income	F	35.08	39.0%
Martinique	MTQ	Other ISO 3166-1 entity	Americas	—	—	—	—
Mauritania	MRT	UN member	Sub-Saharan Africa	Lower middle income	E	56.92	54.9%
Mayotte	MYT	Other ISO 3166-1 entity	Sub-Saharan Africa	—	—	—	—
Micronesia	FSM	UN member	Asia and Pacific	Lower middle income	F	28.58	31.7%
Monaco	MCO	UN member	Western Europe	High income	F	21.42	22.0%
Montserrat	MSR	Other ISO 3166-1 entity	Americas	—	F	16.00	15.9%
Nauru	NRU	UN member	Asia and Pacific	High income	F	42.08	40.2%
New Caledonia	NCL	Other ISO 3166-1 entity	Asia and Pacific	High income	F	7.50	11.0%
Niue	NIU	Other ISO 3166-1 entity	Asia and Pacific	—	F	25.33	18.3%

Norfolk Island	NFK	Other ISO 3166-1 entity	Asia and Pacific	—	—	—	—
North Korea	PRK	UN member	Asia and Pacific	Low income	F	35.00	34.1%
Northern Cyprus	XNC	Not an ISO 3166-1 entitty	Middle East, North Africa & Central Asia	—	—	—	—
Northern Mariana Islands	MNP	Other ISO 3166-1 entity	Asia and Pacific	High income	—	—	—
Oman	OMN	UN member	Middle East, North Africa & Central Asia	High income	E	62.00	54.9%
Palau	PLW	UN member	Asia and Pacific	Upper middle income	F	23.25	31.7%
Palestine	PSE	Other ISO 3166-1 entity	Middle East, North Africa & Central Asia	Upper middle income	F	45.25	42.7%
Papua New Guinea	PNG	UN member	Asia and Pacific	Lower middle income	E	68.50	62.2%
Pitcairn Islands	PCN	Other ISO 3166-1 entity	Asia and Pacific	—	—	—	—
Puerto Rico	PRI	Other ISO 3166-1 entity	Americas	High income	—	—	—
Qatar	QAT	UN member	Middle East, North Africa & Central Asia	High income	E	62.00	56.1%
Republic Srpska	XRS	Not an ISO 3166-1 entitty	Eastern Europe	—	—	—	—
Réunion	REU	Other ISO 3166-1 entity	Sub-Saharan Africa	—	—	—	—
S. Georgia & S. Sandwich	SGS	Other ISO 3166-1 entity	Americas	—	—	—	—
Saint Martin (France)	MAF	Other ISO 3166-1 entity	Americas	High income	—	—	—
Samoa	WSM	UN member	Asia and Pacific	Lower middle income	F	35.08	39.0%
San Marino	SMR	UN member	Western Europe	High income	F	21.42	22.0%
São Tomé & Príncipe	STP	UN member	Sub-Saharan Africa	Lower middle income	F	47.50	48.8%
Seychelles	SYC	UN member	Sub-Saharan Africa	High income	E	75.58	54.9%
Sint Maarten	SXM	Other ISO 3166-1 entity	Americas	High income	—	—	—
Solomon Islands	SLB	UN member	Asia and Pacific	Lower middle income	E	63.50	54.9%
Somalia	SOM	UN member	Sub-Saharan Africa	Low income	F	42.25	46.3%
Somaliland	XSL	Not an ISO 3166-1 entitty	Sub-Saharan Africa	—	F	9.83	11.0%
South Sudan	SSD	UN member	Sub-Saharan Africa	Low income	F	41.25	42.7%
St. Barthélemy	BLM	Other ISO 3166-1 entity	Americas	—	—	—	—
St. Helena	SHN	Other ISO 3166-1 entity	Sub-Saharan Africa	—	—	—	—
St. Kitts & Nevis	KNA	UN member	Americas	High income	E	58.67	48.8%
St. Lucia	LCA	UN member	Americas	Upper middle income	E	87.42	63.4%
St. Pierre & Miquelon	SPM	Other ISO 3166-1 entity	Americas	—	—	—	—
St. Vincent & the Grenadines	VCT	UN member	Americas	Upper middle income	F	48.00	47.6%
Suriname	SUR	UN member	Americas	Upper middle income	F	50.50	53.7%
Svalbard & Jan Mayen	SJM	Other ISO 3166-1 entity	Western Europe	—	—	—	—
Switzerland	CHE	UN member	Western Europe	High income	E	67.08	61.0%
Syria	SYR	UN member	Middle East, North Africa & Central Asia	Low income	F	42.25	42.7%
Taiwan	TWN	Other ISO 3166-1 entity	Asia and Pacific	High income	E	78.25	67.1%
Tanzania	TZA	UN member	Sub-Saharan Africa	Lower middle income	E	71.67	63.4%
Timor-Leste	TLS	UN member	Asia and Pacific	Lower middle income	E	92.58	64.6%
Tokelau	TKL	Other ISO 3166-1 entity	Asia and Pacific	—	F	0.33	2.4%
Tonga	TON	UN member	Asia and Pacific	Upper middle income	F	51.08	43.9%
Turkmenistan	TKM	UN member	Middle East, North Africa & Central Asia	Upper middle income	E	61.00	53.7%
Turks & Caicos	TCA	Other ISO 3166-1 entity	Americas	High income	F	6.67	11.0%
Tuvalu	TUV	UN member	Asia and Pacific	Upper middle income	F	48.75	42.7%
United Arab Emirates	ARE	UN member	Middle East, North Africa & Central Asia	High income	E	73.00	67.1%
US Minor Outlying Islands	UMI	Other ISO 3166-1 entity	Asia and Pacific	—	—	—	—

US Virgin Islands	VIR	Other ISO 3166-1 entity	Americas	High income	—	—	—
Vanuatu	VUT	UN member	Asia and Pacific	Lower middle income	F	33.17	42.7%
Vatican City	VAT	Other ISO 3166-1 entity	Western Europe	—	—	—	—
Venezuela	VEN	UN member	Americas	—	E	69.00	61.0%
Wallis & Futuna	WLF	Other ISO 3166-1 entity	Asia and Pacific	—	F	1.33	3.7%
Western Sahara	ESH	Other ISO 3166-1 entity	Middle East, North Africa & Central Asia	—	—	—	—
Yemen	YEM	UN member	Middle East, North Africa & Central Asia	Low income	F	29.75	37.8%
Zanzibar	EAZ	Not an ISO 3166-1 entitty	Sub-Saharan Africa	—	F	9.83	11.0%

10 Data aggregation and Index calculation

The Index and its sub-components are calculated as aggregations through the tiers of its data model. As per the principles set out in the section on methodological principles (Chapter 3) we have aimed to ensure the Index has a simple and transparent methodology that makes it easy for external users to understand how data flows from the original sources to the overall Index.

10.1 Normalising source data into metrics

The lowest tier of the Index's data model are "metrics", which are translations of original source data. Metrics are calculated by first collating the processed and standardised source data into a single dataset, this dataset is then subset to the countries selected for inclusion in the Index. The source data is then normalised into metrics by scaling the data from 0 to 1, such that 0 represents the lowest performance of group of countries and 1 represents the highest performance:

- The majority of metrics are scaled using "min-max" normalisation where the lowest score in the source data is translated to 0 and the highest score is translated to 1.
- For some metrics the source must be inverted, that is their highest score represents worse performance than their lowest score and thus the highest score is translated to 0 while the lowest score is translated to 1.
- For some metrics both the highest and lowest score in the source data reflects "worse performance" compared to a reference figure. For example, gender parity in employment where the ideal would be a 50:50 split between men and women in the workforce. For these metrics the distance from the reference value is calculated and then this distance is scaled from 0 to 1 such that values closest to the reference point are scored 1 and those furthest from the reference point are scored 0.
- The Bertelsmann Transformation Index (BTI) and the Sustainable Governance Indicators (SGI) are related data projects with differing country coverage that is somewhat influenced by economic development levels. Having compared the data in these sources a differential scaling is applied such that data from the BTI is scaled from 0 to 0.8 and data from the SGI is scaled from 0.3 to 1.

The section on re-scaling and transformation (Appendix D) has more details about how we approach this.

10.2 Aggregation through the data model

After transforming the source data into metrics, subsequent tiers of the data model are calculated as the simple arithmetic mean of its constituent parts:

- Indicators are estimated as the mean of their constituent metrics. For example, the impartial behaviour indicator is calculated as the mean of the four metrics that are aligned to it in the data model and for which there is data (policy implementation is impartial; decisions free of interference; respect for due process; officials impartial in their duties).
- Themes are calculated as the mean of their constituent indicators. For example, the integrity theme is calculated as the mean of the four indicators aligned to it in the data model and for which there is data (impartial behaviour; corruption; sanctions; and, integrity data).
- Domains are calculated as the mean of their constituent themes. For example, the Strategy and Leadership domain is calculated as mean of the four themes aligned to it in the data model and for which there is data (strategic capacity; openness and communications; integrity; and, innovation).
- Finally, the Index is calculated as a mean of the four domains (Strategy & Leadership; Public Policy; National Delivery; and, People and Processes).

At each stage of calculation values are rounded to 2 decimal places to reduce the impact of any spurious precision arising from either the scaling of the original source data or calculation through the various aggregation levels.

The rescaling of data to the 0-1 scale occurs only when converting source data to metrics, the aggregations into the higher tiers of the data model are not rescaled. Therefore, the theoretical minimum of the Index (0) represents the situation that a country being the lowest performing country in all data sources it is present in and the theoretical maximum (1) represents the situation that a country being the highest performing country in all data sources it is present in.

10.3 Handling of missing data

As outlined in the section on data coverage and country selection (Chapter 9), all but one country included in the Index has some degree of missing data. To ensure the methodology remains as simple and transparent as possible it was decided not to make any imputation of missing data. Based on the aggregation methodology this implies that a country's performance in any missing data is equivalent to its average performance in its observed data points.

10.4 Weighting of data

As with the handling of missing data, to maintain the simplicity and transparency of the methodology it was decided not to apply any explicit weighting to the data. While there is no explicit weighting, the structure of the data model creates an implicit weighting structure. However, each country has its own pattern of missing data and therefore the actual weighting of an individual metric on a country's scores will vary depending on what data is does and does not have. The implicit weight for individual metrics varies from 0.179% to 6.25%, read the section on the implied weighting structure (Appendix B) for more details.

11 Sensitivity analysis

The construction of any model and index involves a number of steps that are ultimately based on specific methodological choices, documented in the previous sections of this report. This section analyses the results of 14 tests carried out assess the robustness of these choices.

11.1 Types of sensitivity tests conducted

Table 11.1 outlines the 14 different tests carried out, these have been grouped into four sets:

- Scaling and aggregation tests, which vary how the modelling scales data both in its sources and in its aggregation.
- Coverage tests, which vary the countries included in the modelling.
- Missing data tests, which vary how the modelling treats missing data.
- Weighting tests, which vary how the modelling approaches the weighting of data.

11.2 Analysis of sensitivity tests

For each sensitivity test the relevant methodological change is applied and the Index calculated with all other choices remaining the same as the final model. The test Index scores are then compared to the Index scores in the final model. For each model the following comparison statistics have been calculated, these are presented in Table 11.2, Figure 11.1, Figure 11.2, and Figure 11.3.

- Difference in coverage – the difference in the number of countries included in the final model and the number of countries in the sensitivity test.
- Difference in index score
  - Any difference in score – the number of countries with any difference between their index score calculated in the final model and their index score calculated in the sensitivity test.
  - Difference in score ±0.05 – the number of countries with a difference between their index score in the final model and their index score in the sensitivity test.
  - Mean difference in score – the mean of the differences between index scores in the final model and index scores in the sensitivity test.
  - Mean absolute difference in score - the mean of the absolute differences between index scores in the final model and index scores in the sensitivity test.
- Differences in rank
  - Any difference in score – the number of countries with any difference between their index score calculated in the final model and their index score calculated in the sensitivity test.
  - Difference in score ±0.05 – the number of countries with a difference between their index score in the final model and their index score in the sensitivity test.
  - Mean difference in score – the mean of the differences between index scores in the final model and index scores in the sensitivity test.
  - Mean absolute difference in score - the mean of the absolute differences between index scores in the final model and index scores in the sensitivity test.
- Correlations
  - The Pearson correlation coefficient between the index scores in the final model and the index scores in the sensitivity test.
  - The Kendall rank correlation (tau) correlation between the index scores in the final model and the index scores in the sensitivity test.

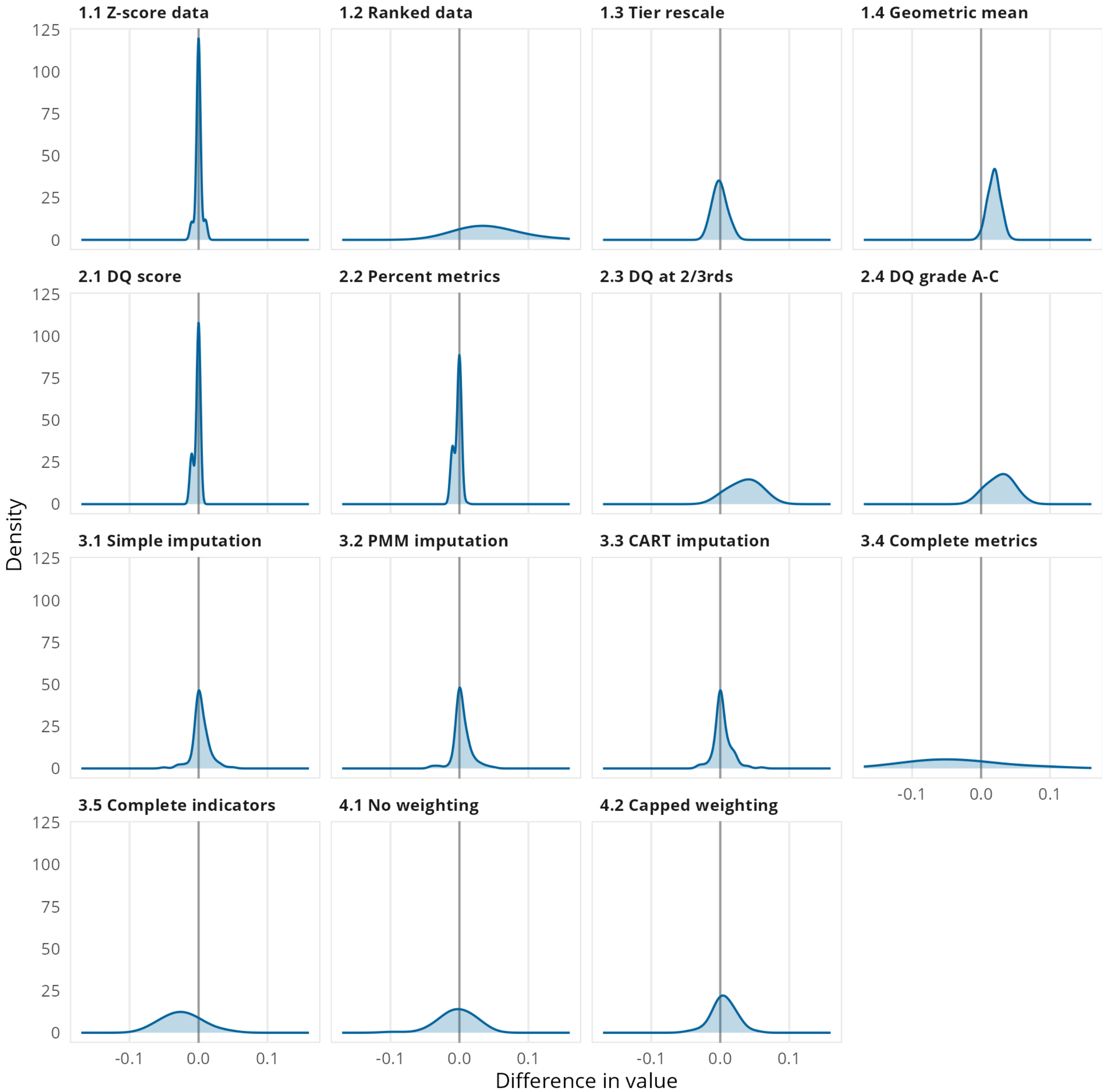
Table 11.1: Summary of sensitivity tests

Sensitivity test	Test description
Scaling aggregation tests	
1.1 Z-score data	Source data is converted to z-scores before its conversion to metrics
1.2 Ranked data	After transformations the data for the metrics are ranked and scaled from 0 to 1
1.3 Tier rescale	Country scores are re-scaled from 0 to 1 after each level of aggregation
1.4 Geometric mean	Aggregation calculated using the geometric mean rather than the arithmetic mean
Coverage tests	
2.1 DC score	Country selection based only on the data coverage score
2.2 Percent metrics	Country selection based only on the percent of metrics
2.3 DC at two-thirds	Country selection using data coverage threshold is harmonised with the threshold for the percent of metrics
2.4 DC grades A-C	Country selection using higher thresholds for both data coverage and percent of metrics
Missing data tests	
3.1 Simple imputation	Missing data replaced using geographic and income group means
3.2 PMM imputation	Missing data replaced using predictive mean matching (PMM)
3.3 CART imputation	Missing data replaced using classification and regression trees (CART)
3.4 Complete metrics	Index calculated using only the 23 metrics with near complete coverage (119-120 countries)
3.5 Complete indicators	Index calculated using only the 17 indicators with near complete coverage (119-120 countries)
Weighting tests	
4.1 No weighting	Index calculated as a simple average of all metrics (i.e. no implicit weighting from the data model)
4.2 Capped weights	Index calculated capping weights implied by the data model

Table 11.2: Results of the sensitivity tests

Sensitivity test	Difference in coverage	Difference in score				Difference in rank				Correlations	
		Any difference in score	Difference of score ≥ ±0.05	Mean difference in score	Mean absolute difference in score	Any difference in rank	Difference in rank ≥ 5	Mean difference in rank	Mean absolute difference in rank	Pearson (r)	Kendall (tau)
Weighting tests											
4.1 No weighting		101	6	0.00	0.02	111	25	-0.2	4.4	0.986	0.919
4.2 Capped weighting		82	0	+0.01	0.01	103	15	-0.3	3.3	0.994	0.949
Missing data tests											
3.1 Simple imputation		53	1	0.00	0.01	79	7	-0.1	1.9	0.997	0.974
3.2 PMM imputation		48	0	0.00	0.01	86	5	-0.3	2.0	0.997	0.970
3.3 CART imputation		52	1	0.00	0.01	82	6	-0.2	2.0	0.997	0.967
3.4 Complete metrics		111	72	-0.04	0.06	112	36	-0.4	8.0	0.948	0.832
3.5 Complete indicators		106	20	-0.02	0.03	113	21	-0.4	4.4	0.987	0.919
Coverage tests											
2.1 DQ score	+7	26	0	0.00	0.00	59	1	0.0	0.9	1.000	0.992
2.2 Percent metrics	+7	34	0	0.00	0.00	71	1	-0.1	0.9	1.000	0.992
2.3 DQ at 2/3rds	-35	75	22	+0.04	0.04	56	1	-0.3	1.4	0.998	0.977
2.4 DQ grade A-C	-35	75	6	+0.03	0.03	56	0	-0.5	1.2	0.998	0.978
Scaling aggregation tests											
1.1 Z-score data		19	0	0.00	0.00	53	1	-0.2	0.8	1.000	0.993
1.2 Ranked data		113	43	+0.04	0.05	111	12	-0.6	4.0	0.988	0.929
1.3 Tier rescale		118	0	0.00	0.01	63	5	-0.4	1.4	0.999	0.984
1.4 Geometric mean		116	0	+0.02	0.02	70	3	-0.3	1.4	0.999	0.984

Figure 11.1: Density plots of difference in index value by sensitivity test



Overall, most tests do not see any substantial variation in score, there is more variation in rank but on average this remains low. All sensitivity tests have strong correlations with the final index scores, whether using Pearson's correlation or the more robust Kendall tau measure.

The largest variations are seen in the tests when restricting to those metrics/indicators with near-complete data, however these still show strong correlation with the final index. The data from these tests are restricted to only a few data sets and a much narrower coverage of the Index's framework, so while they have near complete data they are not necessarily a good representation the Index's intended focus. There is also some volatility when adjusting for the weighting of variables, however it should be noted that this approach also includes imputation of missing data before the re-weighting of data.

Across the 14 tests, the country that ranks first in the final model ranks first in 5 of the tests. Three other countries rank first in other tests (one four times, one 3 times, one twice), all three of these countries rank either 2nd or joint 3rd in the final model. Of the countries that rank first, second or third in the final model, only on 7 instances do they not rank inside the top 3 and in all occasions they rank within the top 10.

Figure 11.2: Density plots of difference in index rank by sensitivity test

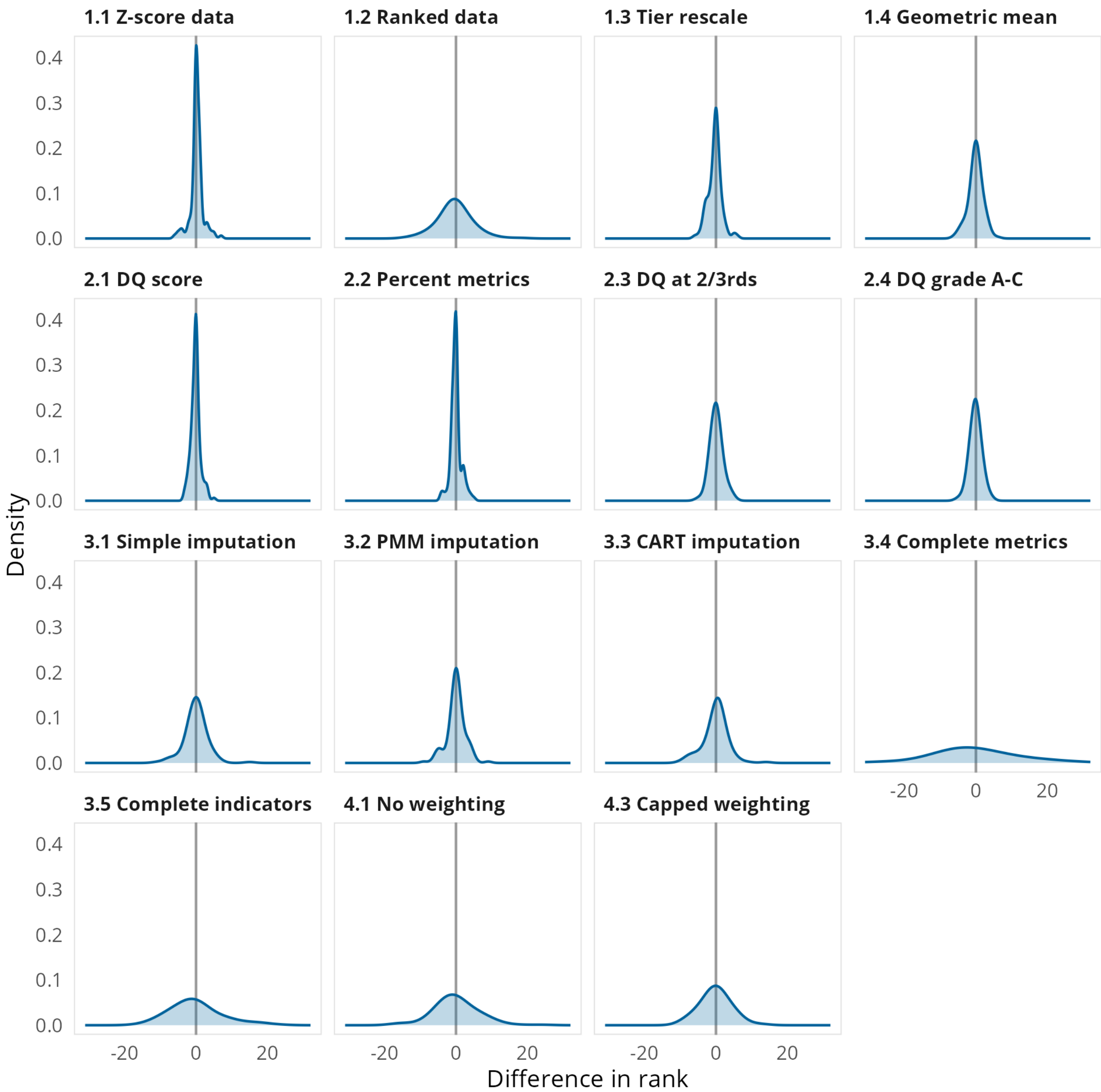
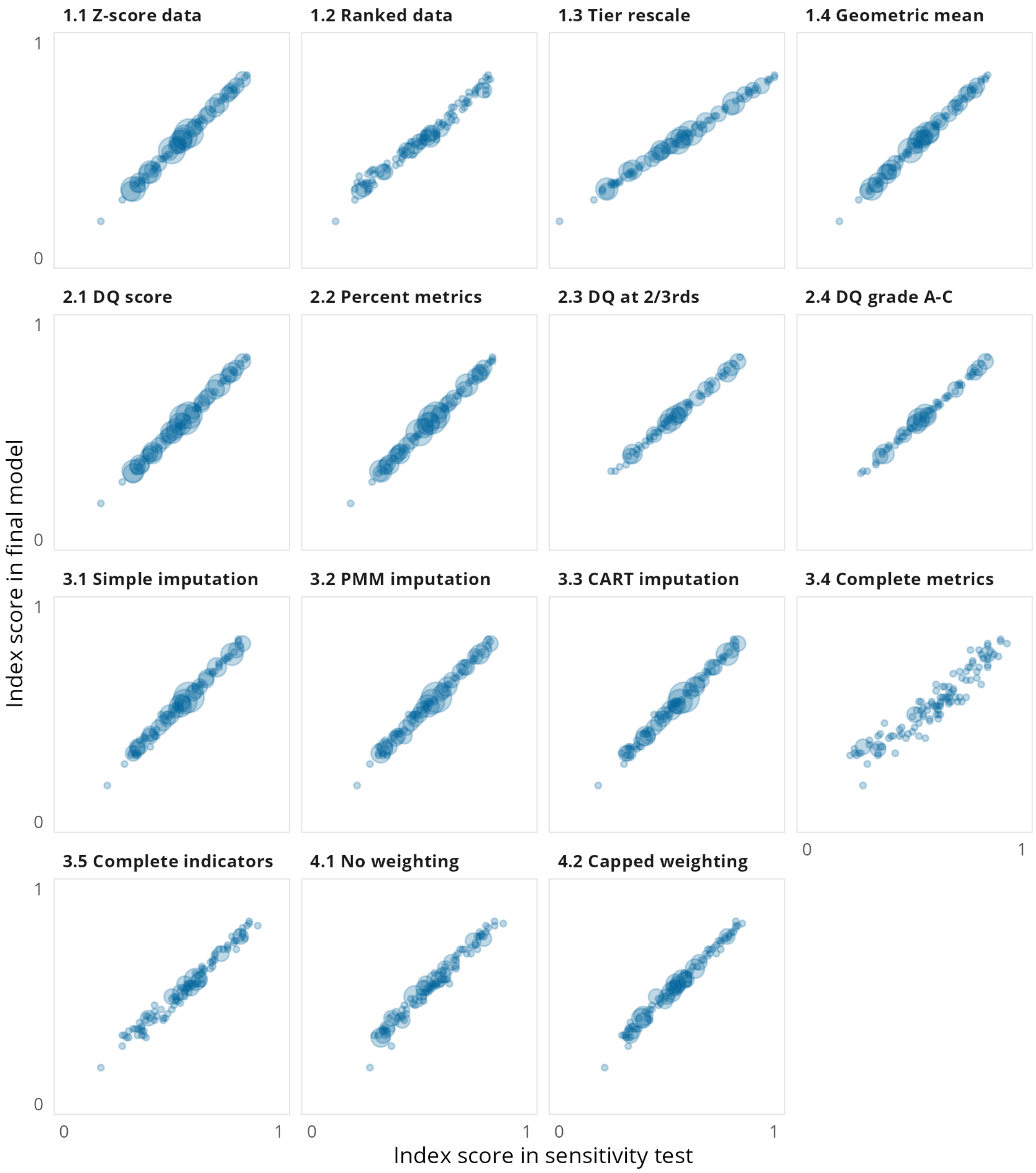


Figure 11.3: Scatter chart of final index scores against index scores in sensitivity tests



11.3 Details of the sensitivity tests

11.3.1 Scaling and aggregation tests

There are four scaling and aggregation tests, two of these adapt the source data and two handle scaling during the aggregation data into the overall Index and its component scores.

Z-score source data (test 1.1)

In this test each variable in the source data is normalised into a z-score (the difference from the variable mean, divided by the standard deviation of the variable) before being input into the model. Each variable has its own original scaling and its own distribution of scores, while the min-max normalisation converts all metrics to a common scale it does not account for the intrinsic distribution of the variables. By normalising all variables into units of standard deviation this test seeks to assess the extent to which the intrinsic measurement scales of each variable influences how countries score.

Ranked data (test 1.2)

In this test after their initial conversion the metric scores are ranked and then these are reconverted from 0 to 1 using min-max normalisation. While it would be preferable to rank the source data, some metrics use distance calculations as part of their transformation, therefore the ranking is performed on the initial calculation of metric scores. As with the z-score test, the aim of this test is to test the impact of the intrinsic measurement scale, using ranks provides non-parametric assessment of this impact.

Tier rescale (test 1.3)

In this test the scores for each tier of the Index's data model are re-scaled from 0 to 1 after their calculation, in the final model this re-scaling only occurs at the calculation of the first tier of the model. This approach was used by the InCiSE 2019 Index. Owing to the original distributions in the source data the distributions of the Index's domains (and themes) are evenly not balanced, the people and process domain has a notably tighter distribution with higher average scores than the distribution of the strategy and leadership domain. Through re-scaling at each tier, this test aims to assess the impact of the attribution of source data to different parts of the data model by ensuring that countries at the relative bottom and top of each tier of the data model are more fairly "rewarded" or "penalised" for their performance.

Geometric mean (test 1.4)

In this test aggregation is made using the geometric mean rather than the arithmetic mean. The arithmetic mean can be strongly influenced by outlier performance, by using the geometric mean this test aims to assess the extent to which country scores are influenced by any outlier performance. The geometric mean is often used to summarise normalised values (https://doi.org/10.1145/5666.5673), however as it is a less intuitive calculation for non-statisticians we have preferred the arithmetic mean. The geometric mean is also intended for use with non-zero positive numbers, to account for when calculating the geometric mean all the metric values had 1 added to them (i.e. the metrics range from 1 to 2), with 1 then subtracted after calculating the final tier so as to return the Index to its intended 0 to 1 range.

11.3.2 Coverage tests

There are four tests which modify the criteria used for country selection. There are two criteria used for country selection, a country's data coverage score and its overall percent of metrics. The data coverage score assesses the spread of a country's available data across the Index's conceptual framework, there are four themes with 10 or more metrics, while there are six themes with only 1 or 2 metrics. Theoretically a country could achieve a high proportion of metrics with data in only a few of the 17 themes, the data coverage score is a method to account for this discrepancy. The overall percent of metrics is used as a second criteria to ensure that countries overall have at least a significant amount of data contributing to their calculations. To qualify for inclusion countries needed a data coverage score that was at least half the maximum possible score (the score if a country has a complete set of data) and had a least two-thirds of metrics overall.

Data coverage (DC) score only (test 2.1)

In this test country selection is based solely on a country's data coverage score. There are 8 countries which pass the threshold for this score but have less than two-thirds of metrics overall (ranging from 54% to 65%):

- Barbados
- Bhutan
- Burundi
- Central African Republic
- Gabon
- Saint Lucia
- Maldives
- Timor-Leste (East Timor)

Percent of metrics only (test 2.2)

In this test country selection is based solely on a country's overall percent of metrics available. There are five countries which have more than two-thirds of metrics (ranging from 67% to 73%) but which do not meet the data coverage threshold:

- Afghanistan
- Cote d'Ivoire
- Democratic Republic of the Congo
- Egypt
- Japan

DC at two-thirds (test 2.3)

In this test the threshold for the data coverage threshold is harmonised with that for the percent of metrics, that is the data coverage threshold is increased from half to two-thirds of the reference score. There are 35 countries which are not included in this test:

- Austria
- Azerbaijan
- Belgium
- Cambodia
- Cameroon
- China
- Eswatini
- Ethiopia
- Gambia
- Germany
- Guinea
- Hong Kong
- Israel
- Italy
- Kosovo
- Lebanon
- Lesotho
- Madagascar
- Malawi
- Mauritius
- Montenegro
- Morocco
- Myanmar
- Namibia
- Nicaragua
- Niger
- North Macedonia
- Norway
- Pakistan
- Rwanda
- Saudi Arabia
- Sierra Leone
- Sudan
- Tajikistan
- Tunisia

DQ grade A-C (test 2.4)

In this test country selection is limited to those countries with data coverage graded from A to C. To help easily explain data coverage countries have also been given a grade from A to F based on the two criteria for inclusion, grades A to D relate to those selected for inclusion with the thresholds set as the upper quartile, median and lower quartile of both the data coverage score and percent of metrics of the 120 countries selected for inclusion. Grades E and F relate to those not selected for inclusion with the threshold being the median of the data coverage scores and percent of metrics of those countries and territories not included. This test excludes those countries included in the main index with comparatively weaker data coverage, in effect testing the assumption that the Index score and data coverage are not correlated. There are 36 countries which are not included in this test:

- Austria
- Azerbaijan
- Belgium
- China
- Eswatini
- Ethiopia
- Gambia
- Germany
- Guinea
- Hong Kong
- Israel
- Italy
- Kosovo
- Lebanon
- Lesotho
- Madagascar
- Malawi
- Mauritius
- Montenegro
- Morocco
- Mozambique
- Myanmar
- Namibia
- Nicaragua
- Niger
- North Macedonia
- Norway
- Pakistan
- Rwanda
- Saudi Arabia
- Sierra Leone
- Singapore
- Sudan
- Tajikistan
- Tunisia
- Zimbabwe

11.3.3 Missing data tests

There are five tests which explore missing data issues. While the criteria for country inclusion takes into account data availability, there is only one country with data for all 86 metrics included in the calculation of the Index. The calculation of the final Index does not employ any imputation to account for missing data, as a result there is an implicit assumption that a country's performance in their missing data is equivalent to their average performance in the data we do have for them. These tests seek to adopt alternative approaches to missing data to test this assumption.

For all but test 3.4 the tests impute missing data for indicators (the second tier of the Index's data model). At the metric tier (the first tier of the data model) there are 10,320 possible data points (86 metrics for 120 countries), of these possible data points 15.6% (1,611) are missing. Metrics are grouped into indicators based on whether they conceptually measure highly similar aspects, for example the corruption indicator is based on three metrics covering public opinion of whether government officials are involved in corruption, expert opinions of whether government officials accept favours for bribes and whether they steal/embezzle public funds. After calculating the indicators we see a significant reduction in the level of missing data, of the 4,320 possible data points (36 indicators for 120 countries) now only 6.7% (290) are missing.

Simple imputation (test 3.1)

In this test missing data is replaced with scores derived from a country's geographic and economic peers. For each indicator the simple arithmetic mean of our six geographic regions and the four World Bank income classification groups are calculated. Where a country has missing data for an indicator this is replaced with a score calculated as half the mean of their geographic region and half the mean of their income group.

Predictive mean matching (PMM) imputation (test 3.2) and classification and regression tree (CART) imputation (test 3.3)

In these two tests missing data is replaced using estimates derived from statistical modelling. In both cases a similar multiple imputation method is followed but the selection algorithm is varied, the mice R package (van Buuren et al) is used to select a set of five candidate values for each missing data point drawn from the observed values of the variable. An average of these values is then used as the replacement for the missing data point. The predictive mean matching (PMM) method uses the correlations of observed data and the pattern of missing data to identify the countries from which to draw the set of candidate values. The classification and regression tree (CART) method uses a supervised learning model to identify the countries from which to draw the set of candidate values.

Complete metrics (test 3.4) and complete indicators (test 3.5)

In these two tests the Index is calculated on a subset of the data model which have complete or near complete coverage for the countries included in the Index. Of the 86 metrics included in the Index, 23 have data for 119 or 120 countries. Of the 36 indicators calculated by the Index, 17 indicators have data for 119 or 120 countries. In each of these tests after calculating results for the given tier the data is subset to only those metrics or indicators and then subsequent tiers are calculated from that data.

11.3.4 Weighting tests

There are two tests which explore approaches to weighting the original source data. In the final model there are no weights directly applied to the data, this results in an implicit weighting model based on the data model, however the actual weight of each metric for each country will vary based on their pattern of missing data.

No weighting (test 4.1)

In this test the index value is calculated as the simple arithmetic mean of all metrics, i.e. the data is not aggregated through the tiers of the Index's data model.

Capped weighting (test 4.2)

In this test weights are applied to the metrics in order to cap the implicit weighting arising from the data model. As outlined in section 11, the implicit weights for a metric vary from 0.017% to 6.25%. There are 17 metrics with an implicit weight of 2% or more, which have an implicit aggregate weight of 52%. This test caps the maximum weight of any individual metric to 2%. For each metric their implicit weight is calculated and then capped, the residual available weight is then redistributed evenly amongst all metrics that were not capped so that the weighting scheme sums to 100%. In order to apply a weighting structure to the data missing data needs to be imputed, for this, the CART imputation (test 3.3) was used, as this imputation is at indicator level the weights for indicators are calculated as the sum of the weights for their constituent metrics.

A Geographic entities reference list and country classifications

One of the critical obstacles to international comparative data analysis is to ensure that data is appropriately matched across sources, to this end the project assigns each country a unique code to match across data sources. Different data providers use different approaches to country naming and coding, sometimes two sources from the same provider can have different approaches.

Country codes and classifications are used within the Blavatnik Index of Public Administration for purely analytical purposes, the inclusion or non-inclusion of a country, territory or geographic entity does not indicate a formal position by the Blavatnik School of Government or the University of Oxford on the legal status of any country, state, territory or geographic feature nor does it indicate an endorsement by the Blavatnik School of Government or the University of Oxford of any claim of sovereignty over any country, state, territory or geographic feature.

A.1 Overview of the geographic entities reference list

The geographic entities reference list has two main purposes: - To provide a code list for the use in the processing of source data, so data from different sources is appropriate matched for each country. - To provide a standard list of names of countries and territories for use in project outputs.

The project uses the ISO 3166-1 alpha-3 standard as the basis for country codes. The alpha-3 codes are preferred as these tend to have a higher degree of visual and linguistic association with the country than the alpha-2 codes.

The processing and assignment of country codes within the source data is made possible by the countrycode R package which provides a comprehensive approach for programmatic assessment and assignment of country codes.

The ISO 3166-1 alpha-3 standard defines codes for 249 geographic entities, including all 193 UN member states and 56 other entities (typically overseas/dependent territories).

In addition to the 249 entities defined in ISO 3166-1, the reference list contains 7 user assigned codes to handle geographic entities that appear in some data sources but do not exist in the ISO 3166-1 standard.

- Kosovo (XKK) – Data for Kosovo appears in many of the data sources. While having a large degree of international recognition, Kosovo is not a member of the United Nations and as such is not included in the ISO 3166-1 standard. The code XK has been widely used as a 2-letter code for Kosovo, however, across the data sources several different 3-letter codes have been observed (KOS, XKK, XKS and XKX). The user-assigned code XKK has been used in our reference list, as in addition to being observed in the source data this is also the 3-letter code used in the Unicode Consortium's Common Locale Data Repository (CLDR; a widely used internet standard for geographic references).
- Northern Cyprus (XNC) – Northern Cyprus does not appear in any sources and is only recognised by Türkyie, it is included in the reference list to aid with processing of cartographic data to produce the basemap in output maps of results. As Northern Cyprus is not included in the ISO 3166-1 standard the user-assigned code XNC has been used.
- Republic Srpska (XRS) – The ISORA data source includes data for the Republic Srpska (one of the two federal entities of Bosnia and Herzegovina). As a sub-national entity, the Republic Srpska does not have an ISO 3166-1 code, the user-assigned code XRS has been used.
- Somaliland (XSL) – The VDM data source includes data for Somaliland separately from the of Somalia. Somaliland is an unrecognised state and is not included in the ISO 3166-1 standard, the user assigned code of XSL has been used.
- State of Palestine, Gaza Strip (XPG) – While the ISO 3166-1 standard includes a code for the State of Palestine (PSE), the VDEM data source provides data separately for the West Bank and the Gaza Strip. The user-assigned code XPG has been used.
- Zanzibar (EAZ) – The VDEM data source includes data for Zanzibar separate to the rest of Tanzania, the ISO 3166-1 standard includes an indeterminate reservation of the code EAZ for Zanzibar and so this has been used.

A.2 Country classifications

To aid in the presentation and analysis of the results two sets of country classifications are used: grouping by geographic regions and grouping by economic classification.

A.2.1 Geographic regions

There are many different approaches for classifying countries into geographic regions, in reviewing the geographic spread of countries included in the Index (see section 5) a bespoke classification has been developed to ensure that each group is of suitable size for comparisons. Our classification assigns countries and territories to one of six groups:

- Americas – 22 countries included in the Index results
- Asia and Pacific – 19 countries included in the Index results
- Eastern Europe – 20 countries included in the Index results
- Middle East, North Africa and Central Asia (MENCA) – 14 countries included in the Index results
- Sub-Saharan Africa – 25 countries included in the Index results
- Western Europe – 16 countries included in the Index results

The classification is based on the regional groupings used by the IMF for their periodic regional economic outlook reports, with some modification. A key reason for selecting the IMF grouping over others as the basis for our classification was its inclusion of a "Middle East and Central Asia" grouping and an "Asia and Pacific" grouping. There are several definitions of which countries comprise "the Middle East", however when using any of these results the number of countries included in the Index results is less than 10. There are only two countries from Oceania included in the Index results (Australia and New Zealand), therefore a grouping which includes these with Asian countries is useful for analytical purposes.

The IMF's regional groupings have been adapted in the following ways:

- Türkiye is included in the IMF's Europe grouping, but for our classification has been included in our Middle East, North Africa and Central Asia (MENCA) grouping. As a transcontinental country Türkiye has significant economic, social and cultural relationships with both Europe and the Middle East. We have included Türkyie within the MENCA group to ensure the grouping has a suitable number of countries and to reflect that Türkiye is included in some definitions of the Middle East.
- Pakistan is included in the IMF's Middle East and Central Asia grouping, but for our classification has been included in the Asia and Pacific grouping. Pakistan's administrative traditions are more similar to other countries in the Indian sub-continent (Bangladesh, India and Sri Lanka) stemming from the legacy of British colonial rule while other central Asian countries share post-communist traditions. Although a Muslim majority nation it is not commonly included in definitions of the Middle East.
- The IMF's Europe grouping has been split into two halves, for simplicity these have been labelled as "Eastern" and "Western" Europe. Without being split, a single Europe grouping would have the largest number of countries represented in the Index (36) and a grouping that is notably larger than the other four groups. For the purposes of our grouping, we define Western Europe as the group of countries that were members of the European Union prior to its 2004 enlargement round, the members of the European Free Trade Association and other European microstates. The remaining countries of Europe are included in our Eastern Europe grouping, of these only the Republic of Cyprus is not a post-Communist state and while geographically in Asia it is a member of the European Union and its administrative traditions are more closely associated with other European countries than those in the MENCA grouping.

A.2.2 Economic classification

For our main economic classification of countries, we use the World Bank's (2023-24)[1] economies by income level classification. This groups countries according to their gross national income per capita into one of four groups:

- High income (GNI per capita > $14,005) – 40 countries included in the Index results
- Upper middle income (GNI per capita $4,516 to $14,005) – 30 countries included in the Index results
- Lower middle income (GNI per capita $1,146 to $4,515) – 31 countries included in the Index results
- Low income (GNI per capita < $1,146) – 12 countries included in the Index results

[1] World Bank, 2024, World Bank Group country classifications by income level for FY24, https://blogs.worldbank.org/en/opendata/new-world-bank-group-country-classifications-income-level-fy24

B Implied weighting structure

As discussed in the section on data aggregation (1), the Index uses a simple mean for aggregating through the levels of the data model to calculate Index and its components. With the exception of the BTI/SGI scaling discussed in the section on re-scaling and transformation (Appendix D), there is no active (re-)weighting of the data at any stage of the Index calculation.

Based on the structure of the data model we can calculate an implied weighting for the individual metrics and sub-components of the model, this is presented in the table below. However, each country has its own pattern of missing data and therefore the actual weighting of an individual metric on a country's scores will vary depending on what data is does and does not have.

The implicit weighting for a metric is calculated as follows:

- Each domain has a within-index weight of 25% as the Index is a simple mean of the four domain scores.
- For the public policy domain each theme has a within-domain weight of 20% as the domain score is calculated as an average of the five theme scores, for the themes in the other three domains their within-domain weight is 25% as these domains have four themes contributing to the calculation of their score.
- Each indicator has a within-theme weight calculated as the inverse of the number of indicators in their theme.
- Each metric has a within-indicator weight calculated as the inverse of the number of metrics in their theme.
- The overall total implied weight for a metric is calculated as the product of its associated within-indicator weight, within-theme weight, within-domain weight and within-index weight.

The implied weights for individual metrics range from 0.179% for the metrics in the data availability indicator to 6.25% for the three metrics which are the sole metrics used to calculate the border services, procurement, and technology & workplaces themes. In total there are 17 metrics with an implied weight of greater than 2%, collectively they have a combined weight of 52.1%. There are 25% metrics with an implied weight of less than 0.25%, collectively they have a combined weight of 8.2%.

Table B.1: Implicit weighting of metrics

Domain[1]	Theme[1]	Indicator[1]	Metric	Metric weight[2]	Overall weighting[3]
Strategy & Leadership (25%)	Strategic capacity (25%)	Prioritisation (100.0%)	Setting strategic priorities	25.00%	1.56%
			Resource efficiency	25.00%	1.56%
			Influence of strategy units	25.00%	1.56%
			Reviewing policy priorities	25.00%	1.56%
	Openness & communications (25%)	Access to information (33.3%)	Right to information	100.00%	2.08%
		Open government (33.3%)	Open government and data portals	50.00%	1.04%
			Published laws	50.00%	1.04%
		Engagement and feedback (33.3%)	Consultation with social actors	25.00%	0.52%
			Consultation with civil society	25.00%	0.52%
			Complaint mechanisms	25.00%	0.52%
			Digital participation methods	25.00%	0.52%
	Trust & integrity (25%)	Impartial behaviour (25.0%)	Policy implementation is impartial	25.00%	0.39%
			Decisions free of interference	25.00%	0.39%
			Respect for due process	25.00%	0.39%
			Officials impartial in their duties	25.00%	0.39%
		Corruption (25.0%)	Officials involved in corruption	33.33%	0.52%
			Officials accept bribes	33.33%	0.52%
			Officials misuse resources	33.33%	0.52%
		Sanctions (25.0%)	Officials prosecuted for abuse	50.00%	0.78%
			Officials sanctioned for misconduct	50.00%	0.78%
		Integrity data (25.0%)	Data for public integrity	100.00%	1.56%
	Innovation (25%)	Innovative approach (100.0%)	Innovative and flexible approach	33.33%	2.08%
			Digital innovation support	33.33%	2.08%
			Innovative tax agency	33.33%	2.08%
Public Policy (25%)	Policy making (25%)	Policy coordination (100.0%)	Coordinating policy proposals	50.00%	3.12%
			Coordinating conflicting objectives	50.00%	3.12%
	Regulation (25%)	Regulatory enforcement (100.0%)	Regulations enforced effectively	50.00%	3.12%
			Regulations enforced properly	50.00%	3.12%
	Crisis & risk management (25%)	Natural hazards (50.0%)	Disaster risk reduction strategies	100.00%	3.12%
		Cybersecurity (50.0%)	Cybersecurity capacity	20.00%	0.62%
			Cybersecurity cooperation	20.00%	0.62%
			Cybersecurity legal measures	20.00%	0.62%
			Cybersecurity organisational measures	20.00%	0.62%
			Cybersecurity technical measures	20.00%	0.62%
	Use of data (25%)	Data capability and governance (25.0%)	Governance of data and statistics	50.00%	0.78%
			Data capability	50.00%	0.78%
		Data availability (25.0%)	Economic statistics availability	14.29%	0.22%
			Company data availability	14.29%	0.22%

			Land data availability	14.29%	0.22%
			Social statistics availability	14.29%	0.22%
			Health data availability	14.29%	0.22%
			Environmental statistics availability	14.29%	0.22%
			Climate data availability	14.29%	0.22%
		Data coverage (25.0%)	Economic statistics coverage	33.33%	0.52%
			Environmental statistics coverage	33.33%	0.52%
			Social statistics coverage	33.33%	0.52%
		Data re-use (25.0%)	Use of statistics in policy documents	100.00%	1.56%
National Delivery (25%)	System oversight (25%)	Policy implementation (100.0%)	Achieve policy objectives	50.00%	3.12%
			Implement policy objectives	50.00%	3.12%
	Digital services (25%)	Digital strategy & policy (33.3%)	Digital government strategy	100.00%	2.08%
		Cross-government infrastructure (33.3%)	Technology and infrastructure	100.00%	2.08%
		End-user services (33.3%)	Digital service portals	100.00%	2.08%
	Tax administration (25%)	Tax organisation (33.3%)	Administrative costs	50.00%	1.04%
			Turnover of tax agency staff	50.00%	1.04%
		Tax compliance (33.3%)	Corporate taxes filed on-time	16.67%	0.35%
			Personal taxes filed on-time	16.67%	0.35%
			Corporate taxes paid on-time	16.67%	0.35%
			Personal taxes paid on-time	16.67%	0.35%
			Time to file taxes	16.67%	0.35%
			Tax debts/arrears	16.67%	0.35%
		Digital tax channels (33.3%)	Corporate taxes filed electronically	25.00%	0.52%
			Personal taxes paid electronically	25.00%	0.52%
			Taxes paid electronically	25.00%	0.52%
			Digital channels for tax advice/help	25.00%	0.52%
	Border services (25%)	Customs (100.0%)	Efficiency of customs processes	100.00%	6.25%
People & Processes (25%)	Diversity & inclusion (25%)	Inclusive recruitment (33.3%)	Inclusive recruitment: gender	20.00%	0.42%
			Inclusive recruitment: location	20.00%	0.42%
			Inclusive recruitment: socio-economic background	20.00%	0.42%
			Inclusive recruitment: social group	20.00%	0.42%
			Inclusive recruitment: political affiliation	20.00%	0.42%
		Gender diversity (33.3%)	Women in public administration (rate)	33.33%	0.69%
			Women in tax agency workforce	33.33%	0.69%
			Ratio of female to male public-sector pay	33.33%	0.69%
		Gender diversity - management (33.3%)	Women in public sector management	50.00%	1.04%
			Women in senior tax agency roles	50.00%	1.04%
	HR management (25%)	HR regime (50.0%)	Recruitment level of entry	50.00%	1.56%
			Security of tenure	50.00%	1.56%
		Meritocratic appointment (50.0%)	Recruitment by merit	33.33%	1.04%
			Recruitment by patronage (QOG)	33.33%	1.04%

			Recruitment not by patronage (VDEM)	33.33%	1.04%
	Procurement (25%)	Procurement data (100.0%)	Public procurement data	100.00%	6.25%
	Technology & workplaces (25%)	Administrative systems (100.0%)	Administrative IT systems	100.00%	6.25%

[1]Percentages in brackets indicate the relative weight for the domain, theme or indicator within its particular part of the Blavatnik Index's data structure. Domains and themes each have a relative weight of 25% since there are four domains that contribute to the overall Index score. Some themes have only one indicator and therefore that indicator has a relative weight of 100%, whereas other themes may be made up of two indicators in which case each indicator has a relative wieght of 50%.
[2]The metric weight column indicates the relative weight of the metric within its parent indicator. This weight is calculated as the simple reciprocal of the total number of metrics that contribute to that indicator (e.g. if an indicator has 3 metrics contributing to it the metric weight is calculated as 1/3, or 33.3%).
[3]The overall weights are those that would apply if a country had data all 82 metrics that are used to calculate the Blavatnik Index of Public Administration 2024. In practice the weighting for each metric varies for each country according to its pattern of missing data. The overall weight is calculated as the product of all the relative weights that apply to a metric throughout the Blavatnik Index data structure. For example the overall metric weight for the metric 'Open government and data portals' is calculated as the domain weight (25%) multiplied by the theme weight (25%) multiplied by the indicator weight (33.3%) multiplied by the metric weight (50%): 25% * 25% * 33.3% * 50% = 1 / (4 * 4 * 3 * 2) = 1 / 96 = 1.04%

C Pre-processing of data sources

All source data files undergo some form of processing to extract them into a common format, see the main section on processing of sources (Chapter 7) for full details, however there are 4 sources which undergo some form of re-calculation of the original data:

- the Global Data Barometer,
- the GovTech Maturity Index,
- the International Survey of Revenue Administration, and
- the Open Data Inventory.

This section summarises the pre-processing we have undertaken of these sources, the source code for this processing is available on Github.

C.1 Global Data Barometer

The D4D.net and ILDA Global Data Barometer attributes its underlying data to both 4 "pillars" (availability, capability, governance, and use and impact) and 7 "modules" (capabilities, climate action, company information, governance, health & covid, land, political integrity, public finance, and public procurement). The re-processing of the Global Data Barometer extracts and aggregates the data for each of the governance and capabilities pillars to create two overall scores for each country, it then aggregates the remaining data on availability and use & impact by module to create seven further scores for each country relating to each module.

The 9 variables extracted from the re-processed Global Data Barometer data cover:

- Climate data availability – sum of the availability scores relating to climate data
- Company data availability and impact – sum of the availability and impact scores relating to company data
- Data capability – sum of the scores relating to data capability measures
- Data governance – sum of the scores relating to data governance measures
- Health data availability – sum of the scores relating to health data
- Integrity data availability and impact – sum of scores relating to the availability and impact of integrity data
- Land data availability and impact – sum of scores relating to the availability of land data
- Procurement data and impact – sum of scores relating to the availability and impact of procurement data
- Public finance data availability – sum of scores relating to finance data availability

C.2 GovTech Maturity Index

The World Bank's GovTech Maturity Index (GTMI) provides a detailed assessment of national governments digital government policies, technologies and digital public services. The high-level results of the GTMI are outputs from a factor analysis of the raw input data, it is therefore not easy to directly identify from the overall results how countries can take action. Furthermore, in addition to its own questionnaire (completed by government officials or World Bank researchers in the case of non-response) the GTMI incorporates some data from other sources which are either already used by the Blavatnik Index of Public Administration (in the case of the ITU's Global Cybersecurity Index) or have been rejected for inclusion (components of the UN's E-Government Development Index). The GTMI data is therefore re-processed to extract its own aggregates of the original raw data.

The variables 7 extracted from the re-processes GTMI data cover: - Administrative IT systems – the existence and nature of IT systems for administration (HR, payroll, financial management and e-procurement) - Back-end technologies and infrastructure – the existence and extent of enabling technologies and frameworks (cloud computing, enterprise architectures, service busses, interoperability frameworks) - Digital methods for public participation – the existence of digital tools to enable public participation and feedback - Digital services and technologies for end-users – the existence of end-user service portals and related technologies (e-payment and digital signatures). - Open government and data portals – the existence and nature of any open government portals and open data repositories - Strategy and policy for digital government – the approach to and policies in support of digital government (strategy documents, institutional structures, open source policies, data privacy laws and institutions, support for skills development) - Support for digital innovations within government – the existence of any policies and practices encouraging digital innovation (innovation strategy, institutional structures, support for digital SMEs).

C.3 International Survey of Revenue Administration

The International Survey of Revenue of Administration (ISORA) is collected and published by the Inter-American Center for Tax Administration [CIAT]; the International Monetary Fund [IMF]; the Intra-European Organization of Tax Administrations [IOTA]; and the Organization for Economic Co-operation and Development [OECD]. The ISORA dataset provides in-depth information about tax collection and operations from 172 tax agencies, the re-processing of the data in order to calculate proportions, coerce categorical data into numerical values, and exclude outlier values.

The 14 variables extracted from the re-processed ISORA data cover: - Administrative costs (% of revenue) – the administrative operating costs of the tax agency as a proportion of net revenue - Corporate tax (% filed on-time) – the number of corporate tax returns filed on time as a proportion of all corporate tax returns - Corporate tax (% paid on-time) – the number of corporate tax returns paid on time as a proportion of all corporate tax returns - Electronic filing: corporate tax – the number of corporate tax returns filed via an electronic method as a proportion of all corporate tax returns - Electronic filing: personal tax – the number of personal tax returns filed via an electronic method as a proportion of all personal tax returns - Electronic payments (%) – the volume of payments made electronically as a proportion of all payments, of the value of payments made electronically as a proportion of the value of all payments (if volume not available). - Personal tax (% filed on-time) – the number of personal tax returns filed on time as a proportion of all personal tax returns. - Personal tax (% paid on-time) – the number of personal tax returns paid on time as a proportion of all personal tax returns - Proportion of all tax officials that are female – the number of officials that are female as a proportion of all officials - Proportion of senior tax officials that are female – the number of senior officials in that are female as a proportion of all senior officials - Proportion of service contacts via digital channels – the number of service contacts via a digital method (online account and chat/digital assistance tools, but not including email) as a proportion of all service contacts (including email) - Tax debt (% of revenue) – the value of tax arrears as a proportion of net revenue - Turnover of tax agency staff – the change in the number of staff in the reporting year as a proportion of the average number of staff for the reporting year - Use of innovative practices and technologies by the tax administration – the extent to which the tax agency is using or putting in place any of 10 practices or technologies (behavioural insights; distributed leger technologies; artificial intelligence or machine learning; cloud computing; data science and analytics; automation; API services; whole of government ID; digital identification technologies; virtual assistants/chatbots).

C.4 Open Data Inventory

The Open Data Watch's Open Data Inventory (ODIN) measures the coverage and openness of official statistics across a several different categories of social, economic and environment statistics. The raw data is processed to calculate a coverage and openness score for each of these three sets of statistics.

The 6 variables extracted from the re-processed ODIN data are: - Economic statistics coverage - Economic statistics openness - Environmental statistics coverage - Environmental statistics openness - Social statistics coverage - Social statistics openness

D Re-scaling and transformation of data

As discussed in the section on data aggregation (1), the metrics for the Blavatnik Index of Public Administration normalised versions of the source data to re-scale them into the range 0 to 1, where 0 represents the lowest performance and 1 represents the highest performance of the group of countries the data relates to. This section provides further details on the re-scaling and transformation methods used.

For the majority of metrics this re-scaling is a simple min-max normalisation, however there are 24 metrics which undergo some other form of transformation.

All the re-scaling operations take place after the source data has been limited to countries identified for inclusion in the Index. Thus, the re-scaled data represents performance relative to countries included in the Index, it does not necessarily represent performance relative to all countries that are included in the original source for that specific metric, nor does it represent the performance relative to any absolute/theoretical scale limits.

D.1 Min-max normalisation

65 of the 89 metrics are re-scaled using min-max normalisation. This normalisation converts the minimum value in the observed data to 0 and the maximum value in the observed data to 1, all other values are re-scaled accordingly.

The metric value for a given country (m_c) is calculated as the difference between the country's score in the source data (x_c) and the minimum score (x_{min}), this difference is then divided by the range of scores in the source data (difference between the maximum score in the source data, x_{max}, and the minimum score in the source data, x_{min}), see equation 1.

$$m_c = \frac{(x_c - x_{min})}{(x_{max} - x_{min})} \tag{1}$$

D.2 Inverted min-max normalisation

For 8 metrics their base scaling is such that the highest score represents worse performance than the lowest score, for these metrics the normalisation is inverted. This is equivalent to conducting a min-max normalisation above but subtracting this score from 1, see equation 2.

$$m_c = 1 - \frac{(x_c - x_{min})}{(x_{max} - x_{min})} \tag{2}$$

D.3 Distance re-scaling

For 4 metrics their re-scaling is based on their distance from a reference point. For these metrics both the highest and lowest scoring countries represent the worst performance and countries closest to the reference point represent the best performance.

- For the two metrics relating to gender equality the reference point is 50%, i.e. the country with the proportion of women in public employment closest to 50% is the country with the highest performance, and vice versa.
- For the metric relating to the gender pay ratio (the ratio of female to male pay in the public sector) the reference point is 1, i.e. the country where female pay is closest to male pay is the country with the highest performance, and vice versa.
- For the metric on staff turnover in tax administrations the reference point is the median value, i.e. both countries with no/very low turnover and countries with very high turnover are low performers. For the gender measures the reference points can be set conceptually – gender equality implies a 50:50 split in the composition of staff and equality of pay between men and women. However, while high and low turnover are both problematic there is no consensus on what an ideal level of turnover should be, therefore the median of the observed data is used.

Table D.1: Average scores across the variables from the BTI and SGI data sources by country inclusion

Data source	Countries only in the BTI	Countries in both the BTI and SGI	Countries only in the SGI
Bertelsmann Transformation Index (BTI)	4.5	6.8	—
Sustainable Governance Indicators (SGI)	—	5.5	6.8

Table D.2: Average minimum and maximum scores across the BTI and SGI data sources for the 15 countries included in both data sources

Data source	Mean minimum score	Mean maximum score
Bertelsmann Transformation Index (BTI)	3.7	9.1
Sustainable Governance Indicators (SGI)	2.9	8.4

First, for each country its distance (d_c) is calculated as the absolute difference between the country's score in the source data (x_c) from the reference value (r), see equation 3. The distances are then used as the inputs to the min-max normalisation to calculate the metric value for the country (m_c), see equation 4.

$$d_c = |x_c - r| \tag{3}$$

$$m_c = \frac{d_c - d_{min}}{d_{max} - d_{min}} \tag{4}$$

D.4 BTI and SGI re-scaling

The Blavatnik Index of Public Administration makes use of two sources from the Bertelsmann Stiftung: the Bertelsmann Transformation Index (BTI) and the Sustainable Governance Indicators (SGI). While these two sources have similar aims, to assess the quality of governance and achievement of public policy goals, and use similar methodologies they have different scopes for country inclusion – the BTI covers countries that were not members of the OECD prior to 1989, while the SGI covers all current OECD and EU members. The SGI was a key data source in the InCiSE 2019 report, in seeking to expand coverage globally, for the Blavatnik Index of Public Administration we have paired 5 of the SGI variables with 5 from the BTI.

153 countries are measured across the two sources: 26 countries are only included in the SGI, 122 countries are only included in the BTI, while 15 countries are included in both the BTI and the SGI. Both the BTI and SGI both ask expert assessors to rate countries on a scale of 1 to 10, and while they have similar aims the questions and intent of the studies is calibrated to the differing development levels of the different sets of countries included in each study. Table D.1 shows the average scores across the 10 variables from the BTI and SGI sources split by country coverage, it shows that the 15 countries included in both datasets score higher in the BTI than the other countries the BTI (6.8 vs 4.8) but in the SGI they score lower than the other countries in the SGI (5.5 vs 6.8).

As implied by the differences in Table D.1, and as shown in Table D.2, across the 15 countries included in both sources their average minimum and maximum scores in the BTI are higher than their average minimum and maximum scores in the SGI. These results suggest that in combining these two different sources there should be some form of calibration of their re-scaling such that data from the SGI is effectively weighted higher than data from the BTI.

Based on the minimum and maximum scores shown in Table D.2, a pragmatic calibration has been adopted such that re-scaled scores from the BTI using min-max normalisation have a range of 0.0 to 0.8 (see equation 5) and re-scaled scores from the SGI using min-max normalisation have a range of 0.3 to 1.0 (equation 6).

BTI normalisation:

$$m_c = \frac{x_c - x_{min}}{(x_{max} - x_{min}} * 0.8 \tag{5}$$

SGI normalisation:

$$m_c = \left(\frac{x_c - x_{min}}{x_{max} - x_{min}} * 0.7\right) + 0.3 \tag{6}$$

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